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MTHM038 - Project Dissertation

Optimising Marketing with Artificial Intelligence: Sentiment and Recommendation Modelling for the Gaming Industry

Dilan Patel (Student ID: 240893718)
Supervisor: Dr Amaranta Membrillo Solis

Contents

Abstract

This research examines the optimisation of marketing, with Artificial Intelligence (AI), for the gaming industry. Stanford Network Analysis Project (SNAP) Amazon reviews for video games were pre-processed and explored. Afterwards, Transformer-Based Sentiment Analysis Models (TBSAMs), Machine Learning Recommendation Models (MLRMs), and a Content-Based Recommendation System (CBRS), were developed. All models were evaluated based on the performance metrics of accuracy, precision, recall, and F1-score, alongside Receiver Operating Characteristic (ROC) analysis. Across the DistilBERT TBSAMs, Round 3 yielded superior performance metrics compared to Rounds 1 and 2. Regarding the MLRMs, a Logistic Regression (LR) classifier outperformed all K-Nearest Neighbours (KNN) options. Lastly, the CBRS consisted of LR predictions, plus suggestions powered by cosine similarity and TF-IDF.

1 Introduction

1.1. Marketing & Artificial Intelligence – Definitions

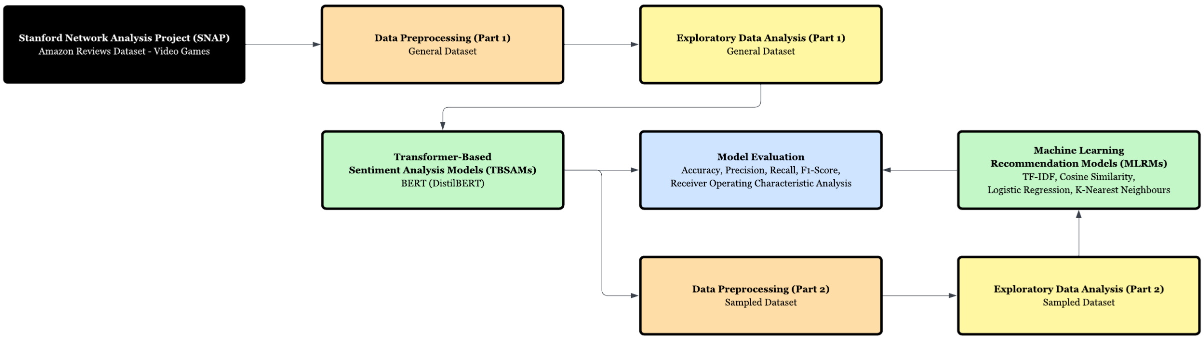
Marketing is the "strategic combination of artistic and scientific elements" and involves the "presentation of products for successful sales, alongside the process of initiating transactions" (Rahman et al., 2024). The sector is responsible for creating, communicating and exchanging valuable offerings for customers, clients and partners. On a societal level, marketing is a collective process, in which individuals and organisations trade goods and services in physical and digital marketplaces. Artificial Intelligence (AI) for marketing is classified into mechanical AI, thinking AI and feeling AI, all of which highlight the software's breadth (Islam et al., 2024); mechanical AI concerns the automation of repetitive tasks, thinking AI regards the processing of unstructured data for strategic decision-making, and feeling AI focuses on understanding emotions towards brands. Furthermore, one could examine the presence and deployment of AI in marketing through the lenses of experience, operations, and competitive intelligence.

Regarding marketing experience, the surge of touchpoints is disrupting traditional models and revolutionising stages of the customer journey (Chen & Prentice, 2024), now contemplated as fluid and non-ordinal (Manthiou & Klaus, 2022). Marketing operations is another department being transformed by AI, providing benchmarks for productivity, value chains, and the granularity of customer interaction (Riyath & Eid, 2025). Using insights from 353 marketing professionals in Sri Lanka, alongside Fuzzy-Set Qualitative Comparative Analysis (FSQCA), Riyath & Eid (2025) found that facilitative environments and effort expectancy are two core conditions impacting large-scale AI adoption for operations. In terms of competitive intelligence, companies can identify potential buyers, strengthen relationships, and maintain engagement, with AI. In fact, propensity marketing is a method of predictive analytics, where brands target specific customer segments based on their unique probabilities of engaging or purchasing. AT&T and Verizon are proponents of AI predictive analytics for forecasting churn, optimising networks, and gauging customer service challenges, whilst IBM Watson supports firms in envisaging patterns between sales variables (Kumar, Ashraf & Nadeem, 2024).

1.2. Marketing & Artificial Intelligence – Project Roadmap & Storage

The research objective was optimising marketing with AI with respect to sentiment and recommendation modelling. Using a Stanford Network Analysis Project (SNAP) Amazon reviews dataset for video games, Transformer-Based Sentiment Analysis Models (TBSAMs) were computed to deconstruct opinions. Following surplus preprocessing, Machine Learning Recommendation Models (MLRMs) were then executed. All models were evaluated using the performance metrics of accuracy, precision, recall, and F1-score, alongside Receiver Operating Characteristic (ROC) analysis. Lastly, the selected MLRM, cosine similarity, and TF-IDF vectorisation of product descriptions, formed a Content-Based Recommendation System (CBRS). Figure A.1.1. depicts the research roadmap. Moreover, the project used Python 3.12, as well as packages including *sklearn*, *pandas*, *matplotlib*, *numpy*, *seaborn*, *statsmodels*, *scipy*, *imblearn*, *transformers*, and *torch*. Jupyter code notebooks (and outputs) are on GitHub, alongside the original datasets. All referenced figures and tables are stored in the Appendix.

Figure A.1.1. Optimising Marketing with Artificial Intelligence (OMAI) – Project Roadmap



1.3. Marketing & Artificial Intelligence – Sentiment

Sentiment is the core of a user-centric environment and the fulfilment of evolving expectations (Nichifor et al., 2023). Termed 'opinion mining', sentiment analysis deciphers the opinions of digital users, expressed through text format, towards products, services, brands and topics (Qorich & El Ouazzani, 2023). The practice applies to R&D feedback cycles, market research projects (Vadla, Suresh & Viswanathan, 2024), and influencer communications for diverse audiences (Suraña-Sánchez & Aramendia-Muneta, 2024). Social proof is inherently connected to sentiment analysis because users tend to source validation and reassurance from reviews (Nichifor et al., 2023). Also, in the context of reviews and sentiment, the word-of-mouth phenomena are measurable through surveys and controlled experiments. Here, several academics emphasise, volume, the idea that greater discussion leads to greater awareness, and valence, which assumes that positive opinions encourage adoption (Dellarocas, Zhang & Awad, 2007). From a customer journey perspective, sentiment analysis relates to the post-purchase phase, where advertisers reinforce the presence of their brand through managing online dialogue. Firms like Autodesk are employing virtual agents to extract the intent and meaning behind post-purchase enquiries (Kietzmann, Paschen & Treen, 2018). Natural Language Processing (NLP) is common technique used for analysing unstructured data, such as online reviews, social media posts and other text, to gain insights into sentiment (Iyelolu et al., 2024). NLP enables AI to deduce the nuances and sentiments behind human language, where companies can adjust their marketing content and campaigns accordingly. For example, Swedbank is just one organisation with an NLP-powered virtual assistant that answers website queries that appear in unstructured format (Kietzmann, Paschen & Treen, 2018).

1.4. Marketing & Artificial Intelligence – Recommendation

Recommendation systems have progressed since e-commerce's inception. Machine Learning (ML) engines were introduced to solve scalability and sparsity troubles surrounding growing data directories and simplistic filtering techniques (Alabi, 2024). For recommendation, AI algorithms can segment customers in profiles, uncover layers of consumer identity to assist 'smart shopping' (Kumar, Ashraf & Nadeem, 2024), simulate user journey scenarios (Iyelolu et al., 2024), and integrate sentiment variables, at a rate faster than human effort. Amazon, Uber and Spotify have all implemented effective AI recommendation systems (Potwora et al., 2024). Holding immense silos of customer behaviour, purchase history and search data, Amazon's algorithms incorporate the prediction of preferences into their recommendation systems, an undeniable factor contributing to the FANG giant's revenue growth. Uber's AI algorithms outline ride recommendations and personalised promotions, where the former is energised through historical travel and geolocational data. Spotify's 'Discover Weekly' playlists are constructed using AI, where profound understanding of audio preferences is instrumental in increasing listener engagement. From a customer journey angle, recommendation relates to the evaluation and consideration phases. Here, advertisers evoke trust in choices sold as optimal (Kietzmann, Paschen & Treen, 2018). Recommending to customers, of high purchase intent and further along the purchase journey, is another archetypal marketing strategy, where recommendation results append lead profiles in real-time, whilst responses to suggestions are often estimated using probabilistic models (Campbell et al., 2020).

1.5. Marketing & Artificial Intelligence – Gaming Industry

There are four distinct areas of the gaming industry where AI is particularly promising. Firstly, microtransactions are shaping player behaviour, where social content, including items that confer status, impacts perceptions of in-game value. Using Partial-Least Squares Structural Equation Modelling (PLS-SEM), and a bootstrapped sample of 10,000 observations, Rita et al. (2024) concluded that social content is significantly associated with microtransactions and impulse buying. Thus, when conducting sentiment analysis and developing recommendation engines, game advertisers should remain aware of the various formats of influential social content. Secondly, the gaming industry is sensitive to culture, where the localisation of sentiment and recommendation marketing systems protects global accessibility. Jiang et al. (2024) analysed 1.2 million player feedback entries across East Asian markets such as China, Japan, and South Korea. By developing an AI model, adaptable to culture, and consisting of multi-layer neural network architecture, the researchers contributed to the sub-discipline of multilingual marketing. Thirdly, the matching of games consumers, and players with other players, is improvable through AI. Chen et al. (2021) presented a matchmaking model that accounts for heterogenous skill levels, loss aversion, pay-to-win strategies, and AI-powered bots. The researchers concluded that, when containing AI bots, game matchmaking policies are more optimisable. Lastly, AI is central to game development itself. Deep Learning (DL) models, including Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), are popular for visual stimuli processing, multi-agent management, and advanced game mechanics (Bonn, 2023). For example, for 'Watch Dogs 2', Ubisoft have manipulated AI for realistic and engaging Non-Playable Character (NPC) dialogues. Overall, AI is commended for in-game analysis, advancing playability, and enhancing user experience (Wang et al., 2020), far beyond marketing and on a holistic product level.

2 Literature

2.1. Literature – Sentiment

Dellarocas, Zhang & Awad (2007) proposed diffusion models to forecast immediate revenue from film releases, using 55,156 reviews from Yahoo, BoxOfficeMojo and Hollywood Reporter. Here, 15,000 Markov Chain Monte Carlo (MCMC) iterations, with 5,000 for a burn-in period, were applied to 4 models. Aftab et al. (2021) conducted sentiment analysis of a Twitter product marketing dataset of 3,000 observations, whilst developing a Recurrent Neural Network (RNN) with a ReLU activation function and a batch size of 256. The academics applied Random Forest (RF) and Naïve Bayes (NB) classification algorithms, achieving 97% accuracy and 94% recall for the former. Hashmi & Yayilgan (2024) compared Support Vector Machine (SVM) and Gradient Boosted Trees (GBT) classifiers to RNN-based Deep Learning and Transformer-Based (BERT, RoBERTa and XLNet) models. Using 4 Amazon reviews datasets, the researchers operated TF-IDF, Word2Vec and FastText embeddings, to account for non-uniform sentiments, where their hybrid FastXCatStack model yielded performance scores in excess of 92%. Complementing Amazon online reviews with synthetic GPT-3.5 data, Vadla, Suresh & Viswanathan (2024) innovated a BERT and a Text-to-Text Transfer Transformer (T5) encoder-decoder model for aspect-based sentiment analysis. Here, the researchers implemented bespoke BERT tokenisation covering a maximum sequence length of 512 and lemmatised formulation, alongside parts-of-speech tagging based on Word2Vec.

Qorich & El Ouazzani (2023) classified 4 million Amazon reviews using a Convolutional Neural Network (CNN) of differentiable layers. The academics fine-tuned the hyperparameters by updating batch sizes between 32 and 64, plus learning rates between 1e^(-3) and 3e^(-3), where convolutions of 1 and 3 obtained superior accuracies. Nichifor et al. (2023) conducted aspect-based opinion mining of 37,616 Google reviews from 85 firms, developing a Mean Score of Reviews (RMS) model based on TF-IDF that yielded an impressive coefficient of determination of 80%. By choosing 1,000 products with Python's *sample*() function, Nguyen et al. (2018) proposed 3 supervised ML and 3 lexicon-based algorithms for 43,620 Amazon consumer reviews, with stratified 5-fold cross-validation for training phases. The ML models outperformed lexicon-based approaches, where SVM, GBT and Logistic Regression (LR) achieved accuracies of 89%, 87%, and 90%, respectively. Tang, Qin & Liu (2015) merged stochastic gradient descent, and a cross-entropy loss function based on backpropagation, with CNN, Long Short-Term Memory (LSTM) and gated RNN (GRNN) models, to encode semantics and produce sentence-level representations. By assigning a convolutional filter output length of 50, the researchers yielded high-performing Conv-GRNN and LSTM-GRNN hybrid algorithms. Iqbal et al. (2022) encoded 1,194,704 Amazon reviews into word embedding vectors before diversifying 2 models. Their initial model, consisting of an LSTM layer, sigmoid activation function, binary cross-entropy loss, and a dense layer, achieved superior accuracy (97%), recall (75%), and F1-score (78%). Liu et al. (2020) proposed a BERT-BiGRU-Softmax DL model, involving hybrid masking and adaptable attention mechanisms. Conducting experiments on over 500,000 product reviews, their proposed model achieved 95.5% accuracy, outperforming other DL approaches such as RNN and BERT-BiLSTM.

Bilal & Almazroi (2022) compared 6 BERT classifiers with 3 Bag of Words (BoW) methods, leveraging a balanced Yelp reviews dataset of 10,000 observations. Their BERT classifiers outperformed their BoW approaches of K-Nearest Neighbours (KNN), NB and SVM, where a sequence length of 320 achieved the highest accuracy (70.7%) and F1-score (71.7%). Conjecturing that the Mexican tourism sector is affected by opinion, Enríquez, Mencía & Segura-Bedmar (2022) advanced pre-trained Transformer models (RoBERTa and GPT2) through HuggingFace's API. Here, data augmentation eased the unbalanced dataset of 30,212 comments, whilst Adam weight decays (of 0.01) were coded. Wang & Goh (2020) opted for Ordinary Least Squares (OLS) regression to test the relationship between components of game experience and user satisfaction. Using 9,333 Amazon game reviews, the researchers found that greater review word count is correlated with negative ratings ($\beta = -0.13$, $p < 0.01$). Danyal et al. (2023) implemented Multinomial NB with grid search optimisation, Localised SVM (LSVM), KNN, Decision Trees (DT), GBT, RF, and LR. Prior to fine-tuning, LSVM achieved the highest accuracy (89.48%) on 50,000 IMDb movie reviews, whilst Multinomial NB achieved 70.69% accuracy on 'Sentiment Polarity Dataset Version 2.0'. Ali et al. (2024) classified 400,000 Amazon product reviews into sentiment 5 categories. In conjunction with K-Fold Cross-Validation, grid search and regularisation methods, BERT, Bi-LSTM and XLNet models were compared, where the former achieved 89% accuracy after 3 training epochs.

2.2. Literature – Recommendation

Bitar & El Alam (2024) investigated whether personalised AI product recommendations impact the tendencies of consumers to purchase. The academics considered intention via Ajzen's 'Theory of Planned Behaviour' and the Technology Acceptance Model (TAM). Upon using multiple regression of 4 independent variables for 300 responses, 'trust' emerged to be positively correlated with purchase intention. Vaishnavi & Kalpana (2023) developed a sentiment-based product recommendation system based on roughly 30,000 records and 15 attributes. The researchers used NLP to extract sentiment features, obtained real-time data through web scraping, applied the Synthetic Minority Over-Sampling Technique (SMOTE) to handle class imbalance, and vectorised reviews with Term Frequency-Inverse Document Frequency (TF-IDF). Here, CatBoost, XGBoost, RF and LR algorithms were applied, where the former achieved brilliant classification accuracy (94%) and precision (99%). Subsequently, their user-based recommendation models achieved lower Root Mean Squared Error (RMSE) compared to item-based counterparts, where the collaborative filtering system was published through Streamlit. Ahuja, Solanki & Nayyar (2019) proposed recommendation systems based on K-Means Clustering and KNN, whilst differentiating between collaborative and content-based filtering. For the former, Within Clustered Sum of Squares (WCSS) was calculated, where RMSE values decreased in line with the number of clusters. Also, the researchers computed user similarity using Pearson correlation coefficients, whilst utilising a Kaggle dataset of 100,000 movie ratings. Patro et al. (2020) proposed a unique Hybrid Action-Related K-Nearest Neighbour (HAR-KNN) recommendation engine, consisting of a behaviour matrix, a feature vector for Euclidean distance computation, and cosine similarity for identifying comparable users. Transferred to 18,501 Amazon reviews, the system involved collaborative filtering, time-awareness, review coverage degree, and product rating predictions.

Using 22,641 fashion reviews, Bellar, Baina & Ballafkih (2024) compared DL models such as CNN, RNN and BiLTSM, whilst evaluating embedding techniques such as BERT, FastText and Word2Vec. Their ensemble model (CNN-RNN-Bi-LSTM) yielded 96.2% accuracy. Wang et al. (2016) sourced Alibaba's 'Mobile Recommendation Algorithm Competition' dataset of 5 million users. Their fusion of LR trained on stochastic gradient ascent, and multiple GBTs, outperformed linear baselines, where F1-scores exhibited improvements between 2-36%. Iwendi et al. (2020) suggested a DL solution for automated diet recommendation. Their LSTM architecture, containing 3 layers, a batch size of 32, a sigmoid activation function, Adam optimisation, and a binary cross-entropy loss function, outperformed other algorithms such as NB, LR, RNN and GRU, in terms of training accuracy (96.5%), precision (98%), recall (99%), and F1-score (99%), for the baseline class. Sharma et al. (2023) created agricultural recommendation system for Indian farmers that struggle with choosing ideal products for rice crops. The research first involved sentiment analysis of rice disease product reviews via K-Means Clustering ($K = 4$) and TextBlob packages. Here, a Radial Basis Function (RBF) with an SVM classifier achieved training and test accuracies, of 94.2% and 95.4%, respectively. Ahmed et al. (2022) applied LR, NB, RF and GBT (XGBoost and CatBoost) algorithms to develop an item-specific collaborative recommendation system based on adjusted cosine similarity and Pearson correlation coefficients. 1.5 million Amazon reviews were converted into a numerical format using $CountVectorizer$, and the engine reached 52% accuracy.

Du et al. (2021) divided a KNN recommendation algorithm into collaborative user (KNN-UserCF) and item (KNN-ItemCF) elements. Here, cross-validation enabled the selection of the number of nearest neighbours, whilst XGBoost, LightGBM and CatBoost algorithms were also explored. Compared to classic approaches, such as Non-Negative Matrix Factorisation (NMF), the former yielded lower Mean Absolute Error (MAE). Choudhury, Mohanty & Jagadev (2021) created a trust-based recommendation system for movies, using a matrix of 610 users, 9724 items, and 100,837 ratings. The researchers installed recommendation algorithms based on a Backpropagation Neural Network (BPNN), Singular Value Decomposition (SVD), and a Deep Neural Network (DNN) alongside 'DNN with Trust' of superior accuracy (83%). Yang et al. (2022)'s Social-Aware Contextualised Graph (SCG) system harnessed time-aware context, context-aware social aggregation, and personalised prediction scores, through the dot-product of the previous two. SCG outperformed quintessential GNN models but decreased in recall percentage as the number of recommendation options grew. The researchers sourced data from Steam Web API, defined co-purchase relations as "the likelihood of two games being bought together", and implemented early stopping procedures. Lei & Chen (2024) selected an SVM recommendation algorithm to handle dimensional records. The researchers trained the SVM model in a real digital media and content platform environment, whilst tuning kernel type, regularisation parameters, and gamma coefficients. Their model attained 85% accuracy, 82% precision, and 80% recall. Lastly, Elahi et al. (2023) extracted sentence embeddings from around 1.5 million Amazon Digital Music reviews of over 450,000 products, using a BERT model with classification head appendment. The researchers deployed Principal Component Analysis (PCA) to reduce dimensionality and optimise the pre-trained BERT model.

3 Data

The research data is from Stanford Network Analysis Project (SNAP), an open-source library developed by Stanford University. Funded since 2004, SNAP is a general-purpose and high-performance system, where the network nodes, edges and attributes are adaptable to multiple processes. Also, developed for large-memory machines, the network balances performance trade-offs, offering over 140 different graphical algorithms (Leskovec & Sosic, 2016). For sentiment and recommendation modelling, a SNAP dataset of Amazon reviews of video games was selected. The entire dataset spans a period between March 2013 to June 1995. Amazon is a world-leading e-commerce platform, offering a breadth of over 12 million products. The utilisation of Amazon reviews is transferable to other media and unstructured text-based formats, such as blogs, tweets, social media posts, and advertisements, which increases the data's credibility (Hashmi & Yayilgan, 2024). Amazon's review system, which is universal irrespective of product categorisation, enables users to rate products from 1 to 5 stars, and to summarise their experiences, in text. Moreover, the firm's reviews and products are connected through scrapeable Amazon Standard Identification Numbers (ASINs) (Nguyen et al., 2018).

3.1. Data – Sentiment

The initial Amazon reviews dataset for video games consisted of 463,669 observations and was loaded using $pandas$. Further product 'categories' and 'descriptions' datasets were merged using '$productid$'. Prior to data preprocessing, the distribution of $score$ was visualised through a bar chart (Figure A.3.1). Here, 139,485 5-star scores, compared to 28,604 and 17,716 1-star and 2-star scores, raised initial suspicions of class imbalance. Outlining keys such as 'None', 'NaN' and 'Null', the first steps of data preprocessing involved removing 129,438 observations with unknown $price$, missing values through $isnull()$, 628 duplicate rows, and 64 missing entries for review $summary$, which led to 333,526 observations. Furthermore, $price$ and $score$ were converted to floats using $.astype(float)$. Then, because $helpfulness$ contained 102,404 NaN values, and inconsistent numbers of upvotes, the variable was dropped entirely.

This research focused on unused and standalone video games. Therefore, hardware-game combinations (e.g. PS5 with Fifa 20) were excluded, but reverses (e.g. Guitar Hero with Guitar Controller) were included. Cheat code discs (e.g. Action Replay) and game e-readers were included, but used games were excluded. Following the extraction of product information from the $categories$ column, case-sensitive search systems, using $str.contains()$ and $isin()$ functions, were developed to identify misclassified products. Here, Tier A and B terms, indicative of a product not being a video game, were defined and weighted according to importance. Tier A words represented broader terms, such as 'Software', 'Movies & TV', 'Books', 'Accessories', 'Hardware', 'Electronics', and 'Platforms', whilst Tier B words embodied subsets of the higher, including 'Chargers', 'Controllers', 'Gamepads', 'Batteries', 'Speakers', 'Gift Cards', 'Remote Controls', 'Headphones', and 'Skins'. Subsequently, the manual removal of misclassified products led to a total of 276,127 reviews spanning 10,428 unique video game titles.

It was decided to create a separate variable for $console$, and focus on games specific to 1 device. Manual searches of the $categories$ column revealed consoles including PC, Mac, PlayStation 2, Game Boy, Xbox, Gamecube, Nintendo 64, and Nintendo Wii. To identify games associated with more than 1 console, console names were stored in a NumPy array, and organised into pairs, where, if the number in $categories$ exceeded 1, 'Multiple' was entered. Also, patterns were outlined to group console types and generations. For example, {'Game Boy', 'Game Boy Advance'} became 'Game Boy Advance', whilst {'Mac', 'PC'} (or separate devices) became 'PC & Mac'. Lastly, $str.contains()$ imputed consoles inferable from titles, whilst games pertaining to 'Multiple' consoles were dropped. After game productisation and console categorisation, outliers for $console$ and $price$ were visualised through bar charts and box plots (Figures B.3.1 and C.3.1). Console categories with reviews lower than 300 were removed, alongside 27,188 outliers for $price$ calculated as $1.5 \times IQR$ from lower and upper bounds. Hence, for Section 4.1, 248,403 reviews were available, consisting of the variables outlined in Table A.3.1.

Valence Aware Dictionary for sEntiment Reasoning (VADER), a rule-based lexicon method for predicting sentiment, was programmed for exploratory and experimental purposes. Upon its NLP inauguration, Hutto & Gilbert (2014) compared VADER to 11 benchmarks, including Linguistic Inquiry & Word Count (LIWC) and Affective Norms for English Words (ANEW), hypothesising that existing lexicons could be progressed with Word-Sense Disambiguation (WSD). Using an annotated corpus of 4,200 social media tweets, and data from Rotten Tomatoes involving 10,605 sentence-level snippets from 2,000 movie reviews, the researchers invented the gold-standard valence-based engine, which is sensitive to expression in digital contexts, and outperformed human raters regarding F1-score (96%). In this research, VADER was applied to all $review$ text, yielding compound polarity scores in the range [-1, 1].

3.2. Data – Recommendation

In Round 3 of Section 4.1, DistilBERT operated as a black-box sentiment analyser. For recommendation modelling, the Round 3 sample and its respective outputs were leveraged. The first preprocessing step for recommendation modelling involved the one-hot encoding of $console$ into a binary format, where 1 indicated the console's presence (0 otherwise). One-hot encoding eliminated ordinality, ensured a suitable numerical format for ML, and provided a detailed representation of console categories as separate entities. In fact, Iwendi et al. (2020) state that backend operations for ML require numerical values, where the researchers also used one-hot encoding through $pd.get\_dummies$. The second encompassed the preparation of game $description$ text for an item-focused and content-based system. Here, $description$ text was lowercased using $str.lower()$, whilst Python's library of pre-defined list of punctuations, $string.punctuation$, helped to reduce noise. Regarding the latter, Iqbal et al. (2022) state that punctuation of "no bearing" accounts for 40 to 50% of review text, where the logic is transferable to product descriptions. Qorich & El Ouazzani (2023) found that stop word removal caused a 2% improvement in accuracy for their CNN model. Thus, to support TF-IDF vectorisation, stop words, such as 'the', 'and' and 'but', were removed from all $description$ text.

Vitally, $price$ was normalised to a standardised range, as ML models are sensitive to feature magnitudes, and therefore, susceptible to bias. $StandardScaler()$ and $price\_scaler.fit\_transform$ normalised the variable to a mean of 0 and a standard deviation of 1.

$$z_{price} = \frac{x_{price} - \mu_{price}}{\sigma_{price}}$$

To complement an item-based recommendation system, the number of reviews per product were aggregated and stored as $review\_counts$. Games of less than 3 reviews were dropped. For the remainder with lower than 10 reviews, Bayesian Adjustment (BA) was programmed on the averages of $score$, DistilBERT positive ($ADPP$) and non-positive probabilities ($ADNPP$), and VADER compound scores ($AVCS$). Emerson & Declerck (2014) also used a Bayesian model, to combine text-based polarity scores and apply outputs to 4 German sentiment lexicons, to develop 'SentiMerge'. Priors, of 0.5 and 3.0, were assigned to average DistilBERT probabilities and average $score$, respectively. Figures A-B.3.2 divulge narrower distributions of average DistilBERT positive probabilities ($AADPP$). The interchangeable BA formula, where k is the review count threshold, N_i is the number of product reviews, and μ is the prior average, is:

$$AADPP_i = \frac{ADPP_i \cdot N_i \cdot \mu \cdot k}{N_i + k} = \frac{ADPP_i \cdot review\_counts_i \cdot 0.5 \cdot 10}{review\_counts_i + 10}$$

Multicollinearity exists when variables are significantly correlated with each other (Shrestha, 2020). For its detection, Variance Inflated Factor (VIF) and scatter plots (Figures C-G.3.2) were computed. VIF measures the increase in variance of a regression coefficient caused by multicollinearity among predictor variables. The VIF equation is $VIF = 1/(1 - R^2)$, where R^2 is the Coefficient of Determination. VIF values close to 1 indicate the independence of predictors, whilst values between 1 and 5 show moderate correlation. Typically, VIF values exceeding 10 signal problematic multicollinearity. Figure C.3.2 displays the weak and acceptable correlation between $price$ and $review\_counts$. Indeed, Table C.3.2 displays VIF values of 154.23697, 134.28386, and 22.48030, for Adjusted Average Score (AAS), Adjusted Average DistilBERT Positive Probability ($AADPP$), and Adjusted Average VADER Compound Score ($AAVCS$), respectively. The scatter plots confirm that the location of positive correlation is amongst these 3 variables. However, because sentiment mechanisms of the recommendation model were solely based on $AADPP$ (not AAS and $AAVSS$), multicollinearity ceased to be a concerning issue.

The recommendation dataset consisted of 1,410 video games, with an average $price$ of 26.55 and $AADPP$ of 0.55529. Redundant variables, such as $review$ and $categories$ were omitted, where text-based sentiments behind the former had already been processed and numerated through the chosen DistilBERT TBSAM. The reduced list of variables is outlined in Table A.3.2, and descriptive statistics are illustrated in Table B.3.2. As stratified sampling was applied for Round 3 of the DistilBERT TBSAM, the distribution of $console$, and binary Adjusted Average Sentiment Score ($AASS$), reflects the original 248,403 observations. For the latter, there were 1,231 and 179 video games regarded as 'Positive' and 'Non-Positive'.

4 Methodology

4.1. Transformer-Based Sentiment Analysis Models (TBSAMs)

4.1.1. TBSAMs – Overview

Respected as a strong Natural Language Processing (NLP) technique (Vaswani et al., 2023), Transformer-Based models were implemented for sentiment analysis. These models provide greater context for text and efficient data parallelisation (Enríquez, Mencía & Segura-Bedmar, 2022), making them applicable to the dataset's *review* column. The first important step for the Transformer-Based Sentiment Analysis Models (TBSAMs) involved sentiment scoring and categorisation, based on the dataset's *score* column, for all 248,403 reviews, as a means of defining a target variable. For Rounds 1 and 2, reviews with scores (stars) of 1.0 to 2.0, 3.0, and 4.0 to 5.0, were classed as 'Negative' (0), 'Neutral' (1), and 'Positive' (2), respectively. For Round 3, reviews with scores of 1.0 to 3.0 and 4.0 to 5.0 were classed as 'Non-Positive' (0) and 'Positive' (1). Thus, despite categorisation, the sentiment labels exhibited an ordinal structure.

4.1.2. TBSAMs – Document-Level Sentiment Analysis (DLSA)

The objective of the TBSAMs was Document-Level Sentiment Analysis (DLSA), which determines the sentiment of an entire document by treating it as a unit of text. Here, each document involved *review* and the review's *summary* (*combined_text*). Overall, DLSA provides a comprehensive picture of the sentiment and overall polarity expressed in longer pieces (Tang, Qin & Liu, 2015). As a result, the TBSAMs were black-box analysers, that generated predicted probabilities for each pre-defined sentiment class that became inputs to the MLRMs defined in Section 4.2. Nevertheless, there are additional SA methods to mention. Aspect-Based Sentiment Analysis (ABSA) focuses on extracting the sentiment associated with aspects or notions across documents. For video game reviews, ABSA could uncover the sentiment towards graphics, storylines, competitiveness, rewards, and characters. Furthermore, Entity-Level Sentiment Analysis (ELSA) uncovers the sentiment towards people or institutions. ELSA complements the analysis of whether a trademark is effectively representing a brand (Nichifor et al., 2023). For the gaming industry, ELSA could decipher the sentiment towards companies such as Sega, Nintendo, Electronic Arts, Ubisoft, Konami, and other powerhouses.

4.1.3. TBSAMs – Bidirectional Encoder Representations from Transformers

A version of the Bidirectional Encoder Representations from Transformers (BERT) model was selected for DLSA based on widespread adulation in NLP literature. Developed by Google in 2018, BERT is based on Transformer architecture, governed by probabilistic frameworks, and pre-trained on a corpus of unlabelled text. Due to bidirectional training, BERT captures the context from both directions of a token and distinguishes between two semantic meanings of the same word, whilst fathoming linguistic nuances and complexities (Hashmi & Yayilgan, 2024). Explanations in Section 4.1.3 reference Devlin et al. (2019) and Vaswani et al. (2017).

4.1.3.A. BERT – Architecture & Pre-Training

BERT consists of a multi-layer bidirectional Transformer encoder architecture. Devlin et al. (2019), the pioneers of BERT, outlined 2 variations of the model, denoting the number of layers as L, hidden size as H, and the quantity of self-attention heads as A. The values of BERT $Base/Large$ were entered as $L = 12/24$, $H = 768/1024$, and $A = 12/16$. The researchers split WordPiece embeddings, of a 30,000-token vocabulary, into token (T), positional (P) and segment embeddings (S). A 'sentence' was defined as an arbitrary span of contiguous text, whilst a 'sequence' was termed as the input token to BERT, perhaps a single sentence or 2 packed together. The first token of each sequence was assigned a special classification label of $[CLS]$. In addition, sentence pairs were combined in a single sequence, with each separated using a $[SEP]$ token. Hence, E_i denotes the input embedding, where $C \in \mathbb{R}^H$ and $T_i \in \mathbb{R}^H$ are the final hidden vectors, of the special $[CLS]$ token and the i^{th} input token, respectively:

$$E_i = T_i + P_i + S_i$$

BERT completed two pre-training tasks. First, Masked Language Modelling (MLM) involved predicting the next word from a string of words. The researchers masked a percentage of input tokens at random and then predicted those masked tokens. In the situation where the i^{th} token was chosen, it was replaced with the $[MASK]$ token 80% of the time, a random token for 10%, and left unchanged for 10%. Hence, for the former, the sentence 'my cat is white' became 'my cat is $[MASK]$', if 'white' was the selected i^{th} token. Next, BERT completed Next Sentence Prediction (NSP), assessing whether the second sentence in a pair genuinely succeeds the first, as a binary classification problem. Consider sentences A and B (for example). 50% of the time, B was defined to be the next sentence following A, whilst the remaining 50% was designated to a random sentence from the corpus. The MLM and NSP formulae are below. For the MLM equation, $\mathcal{M}$ is the set of masked positions, t_i is the true masked token, and E_{Masked} is the input sequence's embedding representation, whereas P is the probability of the true token being at the masked position. For NSP, $IsNext$ and $NotNext$ are binary labels, indicating if the second sentence succeeds the first, or is just a random selection from the corpus, respectively. Hence, P represents the probability of a true successive sentence given the $[CLS]$ embeddings.

$$\mathcal{L}_{MLM} = -\sum_{i \in \mathcal{M}} \log P(t_i \mid E_{Masked}) \qquad \mathcal{L}_{NSP} = -\log P(IsNext \mid CLS)$$

4.1.3.B. BERT – Attention Mechanisms, Networks and Layering

BERT's roots are in Vaswani et al. (2017)'s paper, titled 'Attention Is All You Need'. According to the academics, attention mechanisms are "an integral part of compelling sequence and transduction models". Encoder and decoder stacks guided the formulation of Vaswani et al. (2017)'s attention mechanism. The encoder, comprising $N = 6$ identical layers, mapped an input sequence of symbol representations, $x = (x_1, \dots, x_n)$, to one of continuous representations denoted as $z = (z_1, \dots, z_n)$. Each layer contained two sub-layers, with the relevant output defined as $LayerNorm(x + Sublayer(x))$. The first layer was a multi-head self-attention mechanism, whilst the second was a position-wise connected Feed Forward Network (FFN). For the former, it's useful to define BERT's 'attention function'. The attention function used in BERT is based on the scaled dot-product between query vector Q, value vector V, and key vectors K. Importantly, a $softmax$ function converts a vector of scores, z, into a probability distribution, on a row-wise basis. The term inside the $softmax$ function, generating a matrix of similarity scores between queries and keys, is multiplied by V to yield a weighted sum of values. Queries and keys are of dimensions d_k, whilst values are of dimension d_v. Therefore, the weights, leveraged by BERT to measure attention between pairs of words, originate from these vectors. Due to suspicions that large values of d_k lead to large dot products (and render $softmax$ to be of minute gradient), Vaswani et al. (2017) scaled the function's dot product by $\sqrt{1}/d_k$.

$$softmax(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \qquad Attention(Q,K,V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Hence, 'multi-head' attention means that BERT stacks multiple layers, where each receives the output from the previous layer, enabling the model to attend to different positions. BERT learns multiple attention mechanisms, referred to as 'heads', that operate simultaneously. Also, Vaswani et al. (2017)'s decoder comprised a stack of $N = 6$ identical layers. However, the further insertion of a third sub-layer performed multi-head attention over the output of the encoder stack. Given z, the decoder generated an output sequence, $y_1, \dots, y_m$. Self-attention layers enabled each position to attend to all positions in the decoder, and $softmax$ input values were masked in conjunction with being set to $-\infty$, as to preserve auto-regressive properties. Regarding position-wise FFN, the researchers' encoder and decoder networks contained 2 convolutions of kernel size 1, where input and output model dimensionalities equated to 512, and the intermediate hidden layer size equalled 2048. Vaswani et al. (2017) adjusted their research to account for positional encoding, and the absolute positions of tokens. For positional sine and cosine functions, each dimension of their encoding corresponded to a sinusoid, where wavelengths formed a geometric progression from 2π to $10000 \cdot 2\pi$.

4.1.4. TBSAMs – Distilled BERT (DistilBERT)

Section 4.1.4 cites Sanh et al. (2020). Given Section 4.1.3's mathematics, a pre-trained and distilled version of BERT, titled DistilBERT, formed the basis of the dissertation's TBSAMs. Considering the computational expense of BERT, Sanh et al. (2020) proposed the distilled version, consisting of a 40% size reduction and 97% of the parent's language understanding capabilities. Evaluated through the General Language Understanding Evaluation (GLUE) benchmark, and tested on downstream tasks like IMDb sentiment classification, DistilBERT mirrors BERT's original architecture, but is absent of token-type embeddings. In addition, its number of layers is reduced by a factor of 2. DistilBERT was pre-trained with knowledge distillation, a compression technique. Here, a more compact (student) model was trained to reproduce the behaviour of a larger (teacher) model. Importantly, Sanh et al. (2020) adjusted the original $softmax$ function using a $softmax$ temperature term, denoted by T. Moreover, the researchers retained z_i, which represents the model's score for class i.

$$softmax(z_i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$

DistilBERT's pre-training objective involved minimising a linear combination of distillation, cosine-distance, and MLM losses, denoted as $\mathcal{L}_{KDL}$, $\mathcal{L}_{COS}$, and $\mathcal{L}_{MLM}$, respectively. The first minimised the divergence between student and teacher $softmax$ distributions, encouraging replications of the latter's predictions. The second aligned the hidden state representations of both models. The third, akin to the BERT original, ventured to predict the next word in a sentence. In the triple loss function ($\mathcal{L}_{DB}$) below, t_i and s_i represent probabilities assigned by the teacher (BERT) and student (DistilBERT) models. As $T(x)$ and $S(x)$ are the respective vector outputs, $T(x) = (t_1, \dots, t_n) = softmax(\hat{t}_1, \dots, \hat{t}_n)$ and $S(x) = (s_1, \dots, s_n) = softmax(\hat{s}_1, \dots, s_n)$.

$$\begin{aligned} \mathcal{L}_{DB} &= \mathcal{L}_{KDL} + \mathcal{L}_{COS} + \mathcal{L}_{MLM} \\ &= \sum_i t_i \cdot \log(s_i) + \Big(1 - \cos\big(T(x), S(x)\big)\Big) - \sum_{i \in \mathcal{M}} \log P(t_i \mid E_{Masked}) \end{aligned}$$

4.1.4.A. DistilBERT – Fine-Tuning & Classification

For this research's TBSAMs, there were particular fine-tuning and classification methods of enhancing the pre-trained DistilBERT models. Firstly, a variant of Adam Optimisation ($AdamW$) was included, which involves computing $g = \nabla\mathcal{L}(\theta_{t-1})$ (loss function gradients), updating biased first moment estimates (m_t), updating biased second raw moment estimates (v_t), computing bias-corrected moment estimates ($\widehat{m}_t$, $\hat{v}_t$), and updating parameters through the equation below. Here, θ are the model parameters, α is the adaptive learning rate (step size) for optimisation, and ϵ is small magnitude term to prevent division by 0. λ is a weight decay coefficient, applied to the update formula for consistent regularisation, the prevention of overfitting, and the penalisation of large weights during training (Devlin et al., 2019).

$$\theta_{t+1} = \theta_t - \alpha \cdot \left(\frac{\widehat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda \cdot \theta_t \right)$$

Secondly, to complete downstream DLSA classification, a Cross-Entropy Loss Function (CELF), denoted by $\mathcal{L}_{CELF}$, was incorporated. According to DataCamp, entropy "calculates the degree of disorder in a system". In a data science context, entropy of random variable is "the average event uncertainty or information inherent to possible outcomes". Therefore, cross-entropy is governed by the objective of minimising the error between actual and predicted outcomes. Lower $\mathcal{L}_{CELF}$ indicates stronger model performance, where greater values of p_k reward the given model for making correct predictions. In the equations below, y_k is the true probability of class k, denoted as 1 for correct, and 0 for all others. p_k and z_k are the predicted probability and logit (raw output) values for class k, respectively, whilst K is the total number of classes. Overall, $softmax$ processed the final logits of the DistilBERT TBSAMs to produce probabilities that corresponded to the models' confidence levels in each class label.

$$\mathcal{L}_{CELF} = -\sum_{k=1}^{K} y_k \log(p_k) = -\sum_{k=1}^{K} y_k \log\left(\frac{e^{z_k}}{\sum_{j=1}^{K} e^{z_j}}\right) \qquad \frac{\partial \mathcal{L}_{CELF}}{\partial z_k} = p_k - y_k$$

4.1.4.B. DistilBERT – Model and Round Definitions

Target variables of the TBSAMs were the categorical sentiment labels derived from $score$, and the model input was $combined\_text$. Using $train\_test\_split$ from $sklearn$, the data was portioned into test (20%) and training (80%) segments, where 10% of the former was kept for validation. For reproducibility, a random seed of 1,500 was fixated for each TBSAM. Each round corresponded to a model, symbolising a research decision to alter hyperparameters such as learning rate, weight decay, stratified sample size, and epochs, whilst comparing both different target variables and case-sensitive arrangements. For fine-tuning BERT and the model's relatives, a batch size of 16 (chosen) or 32 is typically endorsed (Bilal & Almazroi, 2022).

The goal of Round 1 was to test the mechanics of DistilBERT through a non-optimised uncased approach, consisting of computationally sparer hyperparameters. Round 1 consisted of 3 epochs, a learning rate of 1e-5, a stratified sample of 5%, and a negative log-likelihood loss function ($nn.NLLLoss$). Round 2 involved 5 epochs, AdamW weight decay of 0.01, a doubled learning rate of 2e-5 (to achieve faster convergence), and a stratified sample of 10%. Here, the DistilBERT version changed to cased. Further, the CELF replaced the Round 1 counterpart, and involved class-weighted adjustment to reduce the effects of class imbalance. Round 3 remained the same as Round 2, aside from an important distinction. To explicitly mitigate class imbalance through the dataset, the target variable became 'Non-Positive' (0) or 'Positive' (1). An Analytics Vidhya publication titled 'Step-by-Step BERT Implementation Guide', coupled with HuggingFace manuals on Transfomers, formed the DistilBERT code.

4.1.4.C. DistilBERT – TBSAM-Specific Data Preprocessing

For Round 1, $combined\_text$ was converted to a consistent and lowercased format, as to complement an uncased version of DistilBERT. Iqbal et al. (2022) state that case-sensitive methods can struggle with determining the polarity of text, and therefore, recommend standardisation. However, the logic is somewhat contentious, especially since capital letters do indicate emotions of surprise, enthusiasm, disdain, and anger. Therefore, Rounds 2 and 3 retained case-sensitivity of $combined\_text$, in conjunction with a cased version of DistilBERT. Furthermore, tokenisation, a technique for diving text streams into phases or chunks of material (Iqbal et al., 2022), was required to initialise vocabulary with individual characters in the language, and to iteratively update the base with frequent combinations of existing words. Because of computational limitations, the model's sequence length was restricted to 128, which nevertheless offered appropriate coverage and efficiency. The tokeniser was loaded through a fast HuggingFace version of $BertTokenizer$, where individual text sequences were transformed into numerical tokens using $tokenizer.batch\_encode\_plus()$.

4.1.5. TBSAMs – Model Assessment

Because of class imbalance, Nguyen et al. (2018) utilised a breadth of performance metrics, a similar approach to this research's assessment of TBSAMs. For Rounds 1 and 2, training, test and validation metrics were interpreted on a 'one-versus-rest' class basis. However, due to the binary nature of the target variable in Round 3, these metrics were construed for the 'Positive' (1) class. Training metrics measured the extent to which the model suited the training data, whereas test metrics regarded the model's generalisation to unseen data. True Positives (TP_i) are the number of instances correctly predicted as positive. False Positives (FP_i) are the number of instances incorrectly predicted as positive. True Negatives (TN_i) are the number of instances correctly predicted as negative. False Negatives (FN_i) are the number of instances incorrectly predicted as negative. Accuracy is the percentage of all correctly predicted instances out of the total number of samples. Precision is the proportion of correctly predicted positive instances out of all predicted positives for class i. Recall, referred to as True Positive Rate (TPR_i), is the fraction of correctly predicted positive instances from all actual positives for class i. The F1-score is the harmonic mean of precision and recall for class i. False Positive Rate (FPR_i) is the ratio of positive cases that were incorrectly identified or classified as class i. Misclassification Rate (MR_i) is the percentage of all incorrect predictions out of the total.

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \qquad Accuracy_i = \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i}$$

$$Recall_i = TPR_i = \frac{TP_i}{TP_i + FN_i} \qquad FPR_i = \frac{FP_i}{FP_i + TN_i} \qquad F1_i = 2 \cdot \frac{Precision_i \times Recall_i}{Precision_i + Recall_i}$$

$$Misclassification\ Rate_i = 1 - \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} = 1 - Accuracy_i$$

$$Macro\ Average = \frac{1}{|C|} \cdot \sum_{i \in C} Metric_i \qquad Weighted\ Average = \frac{1}{N} \cdot \sum_{i \in C} (Metric_i \times Support_i)$$

Receiver Operating Characteristic (ROC) analysis assessed TBSAM performance by visualising the trade-off between TPR_i and FPR_i, and yielding Area Under the Curve (AUC) values. Each point on the ROC curve represented a threshold for classifying predictions, and AUC closer to 1 indicates better performance. For Section 4.1.5, $sklearn.metrics$ and $matplotlib$ were adopted.

4.2. Machine Learning Recommendation Models (MLRMs) and System

4.2.1. MLRMs & RS – Content-Based Recommendation

The objective was an item-specific Content-Based Recommendation System (CBRS). These systems suggest based on similar characteristics between items that users have previously interacted with, alongside the available directory. Content-Based Filtering (CBF) typically compares user and item profiles to predict user-item interactions. For item profiles, the Vector Space Method (VSM) can be used to convert metadata into vectors, where matches are made according to the proximities between aspects. In the Machine Learning Recommendation Models (MLRMs) and CBRS, such profiles were created using each game's $description$. User profile information, of preferences and behaviours, can consist of ratings, queries, purchase histories, upvotes, and reviews. For the MLRMs and CBRS, the user profile data consists of the (adjusted) average probabilistic outputs of explicit feedback ($review$ column), which were generated during Round 3 of the DistilBERT TBSAMs. Moreover, user profile data contained a binary target variable (sentiment label), taking the values of 1 (Positive) or 0 (Non-Positive), derived from the adjusted average of $score$ across all reviews per video game. CBRSs are domain dependent, where aspect-specific keywords, pertaining to game graphics, storylines, genres, characters, competitions, and more, are quantifiable (Bellar, Baina & Ballafkih, 2024).

4.2.2. MLRMs & RS – Term Frequency-Inverse Document Frequency (TF-IDF)

Term Frequency-Inverse Document Frequency (TF-IDF) is a popular NLP technique that quantifies the importance of words in a document whilst searching multiple (a corpus) and is acclaimed by several academics. For example, Li (2024) presented a system comprising ResNET-50 image processing, a DL CNN, and cosine similarity, where TF-IDF for product description embedding enabled the model to achieve 94.72% accuracy. Das (2023) applied TF-IDF on 2 datasets, yielding 93.81% accuracy for the related classifier. TF-IDF vectorised the video game $description$ variable, which served as an input to the MLRMs outlined in Sections 4.2.4 and 4.2.5, alongside the similarity algorithm of Section 4.2.6. Mathematically, Term Frequency (TF_{ij}) measures the frequency of a word in a document, where f_{ij} is the amount that term i appears in product j's description and max_k represents the maximum frequency of term k in that description. Inverse Document Frequency (IDF_i) increases the importance of unique words, attributable to the ratio inside the logarithmic function that approaches 1 as a term appears more. N is the sum of game descriptions, and n_i is the number that contain the term i at least once. Practically, $sklearn$ and $TfidfVectorizer$ transformed descriptions into numerical arrays, which were stored as sparse matrices through $scipy$, for later modelling.

$$TF_{ij} = \frac{f_{ij}}{max_k(f_{kj})} \qquad IDF_i = \log_e\left(\frac{N}{n_i}\right) \qquad TFIDF(w_{ij}) = TF_{ij} \times IDF_i$$

4.2.3. MLRMs & RS – Recommendation Logic

3 distinct tasks were required for the recommendation models and system to operate. Firstly, binary classification involved predicting the probability of a hypothetical user favouring a particular video game, based on the dependent variable of sentiment label and predictors. The dependent variable was synonymous with 1 for 'Recommend' (due to 'Positive' sentiment on average) and 0 for 'Not Recommend' (due to 'Non-Positive' sentiment on average). Secondly, using classification outputs, games were awarded percentage scores, sorted according to probabilistic thresholds of 0.55 for Logistic Regression (LR), and ranked. Thresholds for K-Nearest Neighbours (KNN) proved to be irrelevant due to performance results explained in Section 5.2. Lastly, similarity scoring, powered by cosine similarity, completed the CBRS, which suggested comparable games from a list reduced by the previous two tasks. The hybrid logic was inspired by Wang et. al (2016), who found that merging results of an ensemble of models boosts predictive performance and stated that a linear method of fusion is preferable.

4.2.4. MLRMs & RS – Logistic Regression (LR)

Logistic Regression (LR), also considered a Generalised Linear Model (GLM), is a supervised ML algorithm that completed the classification-based recommendation task. Fascinatingly, Vaishnavi & Kalpana (2023) argue that LR is effective at managing large text datasets and modelling likelihood. There were 4 key assumptions of LR. Firstly, each point was assumed to be independent of others and absent of correlation. Secondly, the target variable was regarded as binary and dichotomous. Thirdly, LR assumed a linear relationship between independent variables and the log odds of the dependent variable. Fourthly, the sample of 1,410 video games was presumed to not contain outliers that distort the estimation of coefficients, and to be sufficiently large to produce stable results (GeeksforGeeks, 2025). For the LR MLRM, the $LogisticRegression$ package from the $sklearn$ library was assembled and adjusted.

4.1.4.A. Logistic Regression (LR) – Variables, Sigmoid Function & Log Odds

For LR, the explanatory variables were $price$, $description$, and $console$, alongside Adjusted Average DistilBERT Positive Probability ($AADPP$) generated during Round 3 of the TBSAMs. As the target variable was sentiment score, the classifier predicted whether the probability that the input (video game) belonged to a specific class (0 for 'Not Recommend' as 'Non-Positive' or 1 for 'Recommend' as 'Positive'). The sigmoid function (σ(z)) is monotonically increasing, taking any real number and mapping it to the range [0, 1], forming an S-shaped curve called the sigmoid or logistic curve. The function tends towards 1 as z reaches ∞, and to 0 as z travels to $-\infty$. Further, 0.5 was designated as the probabilistic threshold for distinguishing between 'Recommend' ('Positive') and 'Not Recommend' ('Non-Positive') class labels.

$$P(Y = Recommend = 1 \mid X) \ = \ \sigma(z) = \frac{1}{1+e^{-z}} \qquad P(Y = NR = 0 \mid X) \ = \ 1 - \sigma(z) = \ 1 - \frac{1}{1+e^{-z}}$$

$$z \ = \ \left(\sum_{i=1}^{n} \beta_i x_i\right) + b \ = \ \beta_0 + \ \beta_1(price) \ + \sum_{j=1}^{k-1} \beta_{2,j}\big(console_j\big) \ + \ \beta_3\,(description) + \ \beta_4(AADPP)$$

x_i is i^{th} observation, $\beta_i = [\beta_1, \beta_2, \beta_3, \beta_4]$ are the model coefficients, and b is the intercept term. z, a linear combination of independent variables, is the dot product of the model's coefficients and intercept. Given the sigmoid function, LR odds, defined as the ratio of the probability of the given event occurring to not occurring, were modelled. Taking the natural logarithm of odds generated log-odds (or logits). Model coefficients represented the change in log-odds of an outcome for a one-unit change in the predictor variable. Hence, the final logistic regression equation, denoted by $p(x)$, was leveraged to find the probabilities of inputs being of said class.

$$\log\left(\frac{p(x)}{1-p(x)}\right) = z = \left(\sum_{i=1}^{n} \beta_i x_i\right) + b \qquad \frac{p(x)}{1-p(x)} = e^{\left(\sum_{i=1}^{n} \beta_i x_i\right)+b}$$

$$p(x) = \frac{e^{\left(\sum_{i=1}^{n} \beta_i x_i\right)+b}}{1 + e^{\left(\sum_{i=1}^{n} \beta_i x_i\right)+b}} = \frac{1}{1 + e^{-\left(\beta_0 + \beta_1(price) + \sum_{j=1}^{k-1} \beta_{2,j}(console_j) + \beta_3(description) + \beta_4(AADPP)\right)}}$$

4.1.4.B. Logistic Regression (LR) – Maximum Likelihood Estimation (MLE)

Parameters of the LR model were estimated through Maximum Likelihood Estimation (MLE), which first required a likelihood function ($\mathcal{L}(\beta) = P(y_i \mid x_i\, ;\, \beta)$). Through $solver\ =\ 'lbfgs'$, the Quasi-Newton Method (QNM) found optimal solutions, and yielded convergence to zero, by minimising the negative of the log-likelihood function ($J(\beta)$). The decision to deploy QNM stemmed from Liu & Nocedal (1989), who introduced the micro-algorithm's limited memory requirements and stellar convergence properties. Unlike gradient descent solvers, QNM is absent of a fixed learning rate, and it computed search direction using an approximation of the inverse Hessian matrix (H_k^{-1}) of second derivatives. Parameter updates $\left(\beta^{(k+1)}\right)$ were performed with 1,000 iterations, and the random seed was set to 16. Furthermore, to avoid overfitting, an L2 regularisation term ($\lambda\|\beta\|_2^2$) was incorporated, to penalise large and more complex weights of the model's loss function; L2 regularisation $\left(\mathcal{L}_{REG}(\beta)\right)$ diminished model coefficients towards zero, and therefore, contributed to more stable and robust results. In the equations expressed below, $\boldsymbol{X}$ is the matrix of input features, $\boldsymbol{y}$ is the vector of observed outcomes, and $\sigma(\boldsymbol{X}\beta)$ represents the vector of predicted probabilities.

$$\mathcal{L}(\beta) = \prod_{i=1}^{n} P(y_i \mid x_i\, ;\, \beta)) = \prod_{i=1}^{n} [\sigma(\beta^T x_i)]^{y_i}\, [1 - \sigma(\beta^T x_i)]^{1-y_i}$$

$$J(\beta) = -\mathcal{L}(\beta) = -\sum_{i=1}^{n} \left[y_i \log\left(\sigma(\beta^T x_i)\right) + (1-y_i)\log\left(1-\sigma(\beta^T x_i)\right)\right] \qquad \nabla J(\beta) = \boldsymbol{X}^T(\sigma(\boldsymbol{X}\beta) - \boldsymbol{y})$$

$$\mathcal{L}_{REG}(\beta) = -\left(\mathcal{L}(\beta) + \lambda\|\beta\|_2^2\right) = -\sum_{i=1}^{n} \left[y_i \log\left(\sigma\left(\beta^T x_i\right)\right) + (1-y_i)\log\left(1-\sigma\left(\beta^T x_i\right)\right)\right] - \lambda\|\beta\|_2^2$$

$$\hat{\beta} = \arg\min_{\beta}[\mathcal{L}_{REG}(\beta)] \qquad \beta^{(k+1)} = \beta^{(k)} - H_k^{-1}\nabla J\left(\beta^{(k)}\right) \qquad \nabla J\left(\hat{\beta}\right) = 0$$

4.2.5. MLRMs & RS – K-Nearest Neighbours (KNN)

Next, LR was compared with varying K-Nearest Neighbours (KNN) classifiers. The classifier's adoption was inspired by Du et al. (2021) and Patro et al. (2020), both mentioned in the literature review (Section 2.2). Here, $KNeighborsClassifier$ from $sklearn$ was deployed. Similar to the LR model, the explanatory variables were $price$, $description$, and $console$, alongside Adjusted Average DistilBERT Positive Probability ($AADPP$). In terms of architecture, KNN found the 'k' closest datapoints (neighbours) to a given input, and for classification, made predictions based on the majority class of $Recommend = Positive = 1$. In general, the KNN algorithm forms no initial assumptions about the data distribution and is therefore considered a non-parametric and instance-based learning method (GeeksforGeeks, 2025).

Step 1 involved fine-tuning the value of k, a hyperparameter which represents the number of nearest neighbours that ought to be considered when solving the classification task. Using $n\_neighbors = k$ and K-Fold Cross Validation (KFCV) (Section 4.2.7), multiple values of k were compared according to Mean Error (ME). Step 2 measured the similarity between the target and training data, where multiple distances were tested for robustness. Step 3 concerned finding the nearest neighbours, classified as datapoints of the smallest distances to the target point. Then, step 4 involved choosing the most frequent category that appeared, a process referred to as 'majority voting'. For step 2, Euclidean and Manhattan distances were judged, by adjusting $p = n$ inside of $metric = "minkowski"$. Euclidean Distance ($EUCL$), corresponding to $p = 2$, is the straight-line distance between two datapoints in Euclidean space. On the other hand, Manhattan Distance ($MANH$), where $p = 1$, measures the total absolute difference between the coordinates of two points. Hence, the respective distance equations are:

$$Distance_{EUCL}(x, X_i) = \sqrt{\sum_{j=1}^{d}\left(x_j - X_{ij}\right)^2} \qquad Distance_{MANH}(x, y) = \sum_{i=1}^{n}|x_i - y_i|$$

4.2.6. MLRMs & RS – Similarity

As mentioned previously, TF-IDF fashioned an item matrix based on video game $description$ text. To complete the CRBS's similarity scoring task, and to enhance the relevance of features, a further measure was required. Due to the objective of comparing items, alongside the method's widespread praise in literature, Cosine Similarity (CS) was used for the CBRS after ML classification and thresholding. Irrespective of document size, Cosine Similarity measures the cosine of the angle between two vectors projected in multi-dimensional space. Here, each dimension corresponded to a word in each video game $description$ document. Cosine Similarity generated values between [-1, 1], where 1 indicates perfect similarity, 0 is orthogonality, and -1 reveals perfect dissimilarity. Items i and j are connected to their respective feature vectors, p and q. For feature k, p_k and q_k are the respective feature values of items i and j, whilst n is the total number of features. Fortunately, $tfidf\_rec.fit\_transform$ and $linear\_kernel$ from $sklearn.metrics$ made cosine similarity scoring a computable reality.

$$Similarity_{CS}(i,j) = \frac{\sum_{k=1}^{n} p_k q_k}{\sqrt{\sum_{k=1}^{n} p_k^2} \cdot \sqrt{\sum_{k=1}^{n} q_k^2}}$$

4.2.7. MLRMs & RS – Model Assessment, Validation and Improvement

For the MLRMs, the same performance metrics were used as in Section 4.1.5 but applied to the context of recommendation instead. For example, precision was the percentage of correctly predicted positive instances out of all predicted positives for the 'Recommend' class. Both LR and KNN output probabilities are continuous numerical values (e.g. 0.67 probability of belonging to the 'Recommend' class) and were used for classification. Therefore, regression-based error performance metrics, such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), were not applicable. Again, Receiver Operating Characteristic Analysis evaluated the trade-off between True Positive Rate and False Positive Rate.

To ensure MLRM predictions were reliable for unseen data, Stratified K-Fold Cross Validation (SKFCV) was adopted using $K = 10$ folds (with 141 games in each) and the $StratifiedKFold$ package from $sklearn.model\_selection$. In general, SKFCV prevents overfitting, reduces variability, and augments model generalisation. Furthermore, stratification ensured the accurate reflection and maintenance of class distribution across all folds. After 10-fold segmentation, the models were trained on $K - 1$ folds and tested on the last. The process was iterated 10 times, and each fold was treated as a testing set at least once. Performance was averaged across the K iterations. The Synthetic Minority Over-Sampling Technique (SMOTE) is when artificial samples are generated for an underrepresented class (e.g. 'Not Recommend' = 'Non-Positive'). Chawla et al. (2002) outline that a combination of oversampling the minority and undersampling the majority classes achieves stronger classifier performance in Naïve Bayes models. By giving a chance for MLRMs to learn minority classes, and being applied to the training data only, SMOTE partly solved the 'Accuracy Paradox'. Here, if there are 1% of products classed as 'Not Recommend', classifiers could predict all data outcomes as 'Positive', yielding a misleading 99% accuracy score in terms of the minority. Using 42 as a random seed, $SMOTE$ from $imblearn.over\_sampling$ was recruited. Finally, to avoid data leakage, SMOTE and SKFCV code was stored in one pipeline, using further $imblearn$ packages.

5 Results

5.1. Results – Transformer-Based Sentiment Analysis Models (TBSAMs)

Full results for the DistilBERT TBSAMs are in Table A.5.1. For training, Round 1 of the DistilBERT TBSAM consisted of 1,754, 1,020 and 7,163 reviews, classed as 'Negative', 'Neutral', and 'Positive'. The model was tested on 219 ('Negative'), 128 ('Neutral') and 896 ('Positive') observations. Round 1 achieved reasonable training, validation and test accuracies of 70.01%, 69.08% and 67.90%. For the same phase, precision, recall and F1-score displayed macro averages of 55.00%, 60.00% and 56.00%, compared to weighted averages of 76.00%, 70.00% and 72.00%. For the testing phase, the metrics were lower, at 51.00%, 54.00% and 52.00% for the macro average, alongside 75.00%, 68.00% and 71.00% for the weighted. The maximum performance score for the 'Neutral' class, across training, validation and test data, was recall, at a mere 43.00% for the second, highlighting the issue of class imbalance. In addition, Round 1 model performance was stronger for the 'Negative' and 'Positive' classes, where AUC values exceeded 85.00% (Figure A.5.1). A Round 2 increase in stratified sample size meant that the training dataset increased to 3,508, 2,039 and 14,326 reviews, for the 'Negative', 'Neutral' and 'Positive' classes. Marginal changes in performance were recorded amidst the increase of epochs to 5, the inclusion of AdamW weight decay at value 0.01, the switch to a CELF, a doubled learning rate of 2e-5, and the launch of a cased version of DistilBERT. Here, global training, validation and test accuracies were 68.62%, 68.32% and 69.30%, respectively. Macro and weighted averages of precision, recall and F1-score, for all phases and classes, were proximate to Round 1 equivalents. For example, these macro and weighted test metrics were 53.00%, 59.00% and 54.00%, and 76.00%, 69.00% and 71.00%.

Round 3 retained a stratified sample size of 10%, and the same architecture as Round 2, aside from the merge of 'Neutral' and 'Negative' reviews. The measure to mitigate class imbalance increased global training, validation and test accuracies to 77.24%, 78.58% and 76.90%. Further, macro and weighted averages of training precision, recall and F1-score were 73.00%, 76.00% and 74.00%, alongside 80.00%, 77.00% and 78.00%, respectively. The corresponding metrics for validation revealed macro averages of 74.00%, 78.00% and 76.00%, in conjunction with weighted averages of 81.00%, 79.00% and 79.00%. For test data, performance scores were also promising, yielding macro and weighted averages of 73.00%, 77.00% and 74.00%, together with 80.00%, 77.00% and 78.00%, respectively. AUC totals of 84.66% (training), 85.99% (validation) and 84.74% (test) are illustrated in the ROC curve (Figure C.5.1) for the 'Positive' class. Round 3 training and validation losses are plotted in Figure D.5.1 and were the lowest across the epochs. These losses started at 0.646 and 0.589, before converging towards 0.502 and 0.472. The TBSAM of Round 3 also generated DistilBERT predictions of sentiment label, where the respective densities are shown in Figures E-F.5.1. Figure G.5.1 indicates that DistilBERT classified a lower number of 'Positive' reviews (12,607) and a higher quantity of 'Non-Positive' reviews (7,266) compared to labels derived from raw *score*. To reiterate, the stratified sample generated in the Round 3 DistilBERT TBSAM became the basis of the MLRMs and the CBRS, where exploration of the sampled 1,410 games is in Section 3.2.

5.2. Results – Machine Learning Recommendation Models (MLRMs)

For a unit increase in a Logistic Regression (LR) predictor (x_i), odds are multiplied by e^{β_i}. Hence, the coefficients in Tables B-C.5.3, regarded as fixed parameters, encompass the change in the log-odds of the outcome for a (scaled) unit change in a predictor variable. It was found that a unit increase in $price$ increases the odds, of a game being a positive recommendation, by 34.443%. Coefficients for $console$ were interpreted with respect to Sega Genesis, the base category. Here, the odds of a game being a positive recommendation decrease by 17.777%, 57.042%, 41.107%, 28.862%, 22.742% and 23.152%, when belonging to PC & Mac, Sega Dreamcast, Game Boy Color, Nintendo 64, Nintendo Wii and PlayStation $console$ categories, respectively. In contrast, increases of 38.241%, 238.773%, 15.359%, 10.782%, and 105.285%, are inferable for Game Boy Advance, GameCube, Nintendo DS, PlayStation 2 and Xbox games. Coefficients of $description$ words were also extracted. For a scaled unit increase in the TF-IDF scores of the words $players$, $story$, $monsters$, $characters$, and $everquest$, the odds of a game being a positive recommendation rise by 112.659%, 149.540%, 93.193%, 73.454%, and 59.437%. Full performance results for the LR MLRM are in Table A.5.2. The LR MLRM yielded global training and test accuracies of 96.90% and 87.23%. In addition, the weighted averages of test precision, recall and F1-score, were 86.00%, 87.00% and 87.00%, signalling adaptation to unseen data. Figure A.5.2 demonstrates the AUC scores for the 'Recommend' ('Positive') class. Lastly, SKFCV generated average accuracy of 85.82% across all 10 folds.

Prior to K-Nearest Neighbours (KNN) modelling, Figure B.5.2 revealed an increase in Mean Error (ME) in conjunction with a rise in the number of neighbours (K). Due to plateaus in ME for $K \geq 9$, $K = 1-8$ neighbours were modelled using the two different distance measures. For Manhattan distance, 1, 2, 3 and 5 neighbours generated global test accuracies of 81.56%, 75.53%, 76.24%, and 67.73%, respectively. However, for the remaining neighbour choices, test accuracies failed to exceed 59.00%. Moreover, macro and weighted averages of test performance metrics, for 1, 2 and 3 neighbours, were distorted by the majority class. For example, the weighted averages of test precision, recall and F1-score were 78.00%, 76.00% and 77.00% for 3 neighbours, where scores of 87.00%, 85.00% and 86.00% for 'Recommend' (P) dominate 12.00%, 14.00% and 13.00% for the 'Not Recommend' (NP) class. Moreover, for Euclidean distance, global test accuracies were consistently poor, ranging from 12.77% for 8 neighbours, to 28.37% for 1 neighbour. In addition, recall and F1-score for the 'Recommend' (P) class remained elusive across all KNN models. For 1, 3, 5 and 7 neighbours, recall amounted to 22.00%, 8.00%, 4.00% and 2.00%, whilst F1-score revealed 33.00%, 14.00%, 8.00%, and 4.00%, respectively. The differences in AUC values across Manhattan and Euclidean distances are comparable through Figures C-D.5.2. Due to the evidently inferior performance of KNN, it was decided to base the probabilistic element of the CBRS on the outputs of the LR MLRM alone. Therefore, the KNN classification task served more as an option for comparison rather than a direct input in the final recommendation system.

5.3. Results – Content-Based Recommendation System (CBRS)

Here, ‘logistic probability’ is the percentage probability that a game would be a ‘Positive’ recommendation, considering the model’s coefficients. This corresponds to the likelihood of a user viewing the game recommendation as ‘Positive’. For each product, respective observation parameters were substituted into the model’s fitted equation. Then, a sigmoid function predicted the arrangeable probabilities of $Y = 1$. Tables A-C.5.3 outline CBRS examples specific to logistic probabilities. The 5 highest ranked games were ‘Mario Kart: Super Circuit’ on Game Boy Advance (99.725%), ‘Lego Star Wars II: The Original Trilogy’ on PlayStation 2 (99.413%), ‘Mario Party 7’ on GameCube (99.344%), ‘Mario Kart DS’ on Nintendo DS (99.322%), and ‘Halo: Combat Evolved’ on Xbox (99.316%). The lowest 5 exceeding the threshold were ‘Dark Messiah of Might & Magic Elements’ for Xbox 360 (55.101%), together with ‘Resident Evil 4’ (55.056%), ‘Left 4 Dead’ (55.149%), ‘Majestic Chess’ (55.166%), and ‘High Heat Baseball 2003’ (55.173%), all for PC & Mac. Also, the CBRS is adjustable for $console$. For Xbox 360, the highest logistic probabilities are for ‘Devil May Cry 4’ (97.732%), ‘Call of Duty 4: Modern Warfare’ (97.520%), ‘The Orange Box’ (96.233%), ‘Star Wars: The Force Unleashed’ (95.926%), and ‘Burnout Paradise’ (95.697%). Following ‘Mario Kart: Super Circuit’, the highest ranked games for Game Boy Advance are ‘Tony Hawk’s Pro Skater 2’ (98.820%), ‘Super Mario Advance’ (98.116%) and ‘Kirby Nightmare in Dream Land’ (97.815%). It’s also possible to include a $price$ filter. For games priced at 20 or above, some notable mentions are ‘CTR: Crash Team Racing’ for PlayStation (99.315%), ‘Nintendogs Lab & Friends’ for Nintendo DS (99.071%), and ‘Wario World’ for GameCube (99.039%).

Table D.5.3 is a sampled representation of the CBRS, consisting of 5 hypothetical scenarios, ranking based on cosine similarity score, and logistic probabilities. Scenario 1 asked for video games similar to ‘Super Mario 64 DS’, in the $price$ range of 5 to 30 and for Nintendo DS. The top 5 suggestions were ‘Mario Kart DS’, ‘Nanostray’, ‘Touchmaster’, ‘Meteos’, and ‘ Castlevania Dawn of Sorrow’, displaying similarity scores of 0.305531, 0.191583, 0.187996, 0.141974, and 0.134284, respectively. Scenario 2 requested titles similar to ‘Virtua Tennis 3’, removing $console$ and $price$ filters. The CBRS advocated for Sega Dreamcast’s ‘Virtua Tennis’ (0.485578), GameCube’s ‘Mario Power Tennis’ (0.222006), and Xbox 360’s ‘Table Tennis’ (0.167205). Scenario 3 called for PlayStation 2 games, with prices between 5 and 30, similar to the infamous ‘Grand Theft Auto 3’. The top 5 recommendations of ‘Greatest Hits: Grand Theft Auto III’ (0.525864), ‘True Crime Streets of LA’ (0.193973), ‘Grand Theft Auto: The Trilogy’ (0.187680), Grand Theft Auto: Vice City (0.181681), and ‘The Getaway’ (0.109644), make intuitive sense from a crime genre perspective. Scenario 4 stipulated PC & Mac games of an identical $price$ filter, for ‘Rollercoaster Tycoon 3’. Although the CBRS proposed earlier versions of the game like ‘Rollercoaster Tycoon Gold Edition’ (0.394297), scores of 0.653565 for ‘Rollercoaster Tycoon 3: Gold Compilation Pack’, and 0.375407 for ‘Rollercoaster Tycoon 3: Soaked! Expansion’, indicate the value of using recommendations to upsell products. Lastly, Scenario 5 involved an unfiltered command for titles similar to ‘Battlefield: Vietnam’, which offered ‘Battlefield 2: Modern Combat’ (0.20294), ‘Battlefield 1942: Deluxe Edition’ (0.195767), and ‘Joint Operations: Typhoon Rising’ (0.187984), as plausible options.

6 Discussion

6.1. Discussion – Academic Considerations

Regarding this research, there are considerations for academics. Class imbalance is a recurring problem in sentiment analysis, where performance metrics of majority classes outweigh their counterparts. For the 'Neutral' class, the Round 1 DistilBERT TBSAM yielded test precision, recall and F1-scores of 18%, 29% and 22%, respectively. The inferior class-based performance is attributable to limited observations (128) for 'Neutral' compared to 'Positive' (896). For the Logistic Regression MLRM, the test portion consisted of 36 games classed as 'Non-Positive', and 246 as 'Positive'. Here, the imbalance translated to lower precision, recall and F1-scores for the former (50%, 39% and 44%) compared to the latter (91%, 94% and 93%). Several academics have sought to solve such problems. Suhaeni & Yong (2023) considered a Coursera online course review dataset of high imbalance, and undertook synthetic review generation, to fine-tune a Davinci model from GPT-3, whilst conducting sentiment classification through 9 models. Moreno-Garcia & Elyan (2021) applied unsupervised K-Means and Density-Based Spatial Clustering algorithms to 2 sentiment datasets. The researchers found that clustering could reduce the dominance of majority categories without causing substantial information leakages. Analytics Vidhya outline more challenges of NLP for sentiment analysis. Sarcasm can reverse the linguistic sentiment of a sentiment, and the connotations of specific (slang) words change depending on cultural context (e.g. 'sick' for Londoners). Double negatives like 'not bad' are certainly conceivable, whilst decoding emoji-filled text is an emerging demand.

For recommendation modelling, academics ought to consider the omni-technological nature of these systems in reality. Models must be equipped to process real-time data, and to enable deep personalisation, all whilst being connected to other marketing technologies (Alabi, 2024). In fact, Augmented Reality (AR), Virtual Reality (VR), the Internet of Things (IoT), and voice search, could power immersive and multimodal shopping experiences. Furthermore, although academic research tends to be disparate from organisational settings, hypothetical recommendation models should be developed with measurement in mind. Both quantitative and qualitative Key Performance Indicators (KPIs) of success offer a complete view of a recommendation system's effectiveness. For example, Conversion Rate (CR) could be defined as the percentage of customers purchasing a product after viewing a recommendation. Also, Net Promoter Score (NPS), a market research metric measuring the likelihood of respondents recommending a brand or product, is connected to sentiment analysis. Nevertheless, the cold-start problem is the leading challenge of most recommendation systems. The issue emerges when new users or products are added to recommendation systems, where there exists no prior information of either. For users, data on past purchasing or viewing behaviour could be limited, making it difficult to personalise recommendations based on collaborative filtering. Here, users could be reluctant to provide information for reasons expressed in section 6.3. Interestingly, Herce-Zelaya et al. (2020) first created consumer behavioural profiles using Twitter data before making predictions with ML. However, this research explored content-based methods. These are still prone to the cold-start problem, where new products could be devoid of the ratings and reviews necessary for meaningful analysis (Fayyaz et al., 2020).

6.2. Discussion – Commercial Considerations

The first consideration for commercial professionals is data management strategy. The foundation for effective sentiment and recommendation models is the accurate integration of diverse data sources from various touchpoints (Alabi, 2024). Certain firms have heavy data silos, where structured and unstructured data is scattered across platforms and departments. Therefore, these organisations should invest in robust data infrastructure, whilst preserving the completeness, timeliness and relevance of information engineering (Iyelolu et al., 2024). Such companies could consider the DIKW Pyramid, a framework describing the hierarchical relationship between Data (D), Information (I), Knowledge (K) and Wisdom (W); the base of the pyramid indicates that firms must possess foundational abilities for data processing, first and foremost (Campbell et al., 2020). The second commercial consideration is human capital strategy. Firms looking to leverage AI must plan for technical data-focused and non-technical support functions. Critically, to develop optimal sentiment and recommendation models, data scientists require deep understanding of a company's circumstances, markets, objectives, and customers. A gradual transition to a culture of continuous learning, achieved through both training programmes and interdepartmental collaboration, could minimise the pressures of new tools and technologies (Iyelolu et al., 2024). Through research with 18 domain experts, Keegan, Dennehy & Naudé (2022) found that AI-powered B2B marketing systems are influenced by the division of labour, supporting the argument for workforce planning. Also, despite concerns of job automation that disseminate through the grapevine, Huang & Rust (2022a) emphasise the cooperative relationship between AI and human intelligence.

Research & Development (R&D) strategy applies to companies with available capital. R&D into AI-powered marketing technologies could increase Return on Investment (ROI) (Rahman et al., 2024). Generative AI tools formed on deep learning, such as GPT-4 and DALL-E, could create humanistic campaign content (Dimitrieska, 2024). Programmatic advertising unites advertisers, publishers, Demand-Side Platforms (DSPs) and Data Management Platforms (DMPs), in an ecosystem of media inventory, space acquisition, and biddable systems. In particular, DMPs are enhancing audience profiles using ML algorithms (Kumar, Ashraf & Nadeem, 2024). Enhanced with AI, chatbots and virtual assistants are intelligent agent technologies capable of altering customer service and social community management. Overall, companies venturing into AI R&D could benefit from Dynamic Capability Theory (DCT), a framework that supports the adaptation, renewal and configuration of resources through a Resource-Based View (RBV) (Kumar, Ashraf & Nadeem, 2024). Lastly, commercial leaders in the gaming industry must consider broader innovation strategies and landscapes. Park & Larson (2022) used Social Network Analysis (SNA) and a Sectoral Innovation Systems (SIS) framework to examine the rates of network evolution and company behavioural change in Korea's gaming industry. The researchers uncovered a sparse firm network, with an average density between 0.002 and 0.006, where 1,813 firms had 2,210 business transactional links between 2008 and 2017, indicating room for improvement. Xi et al. (2022) found that innovation efficiency in China's video game industry depends on capital investment, market size, and political support. Therefore, firms must anticipate both micro-financial and macroeconomic factors when deploying AI-driven marketing as part of an innovation strategy.

6.3. Discussion – Legal & Ethical Considerations

Algorithmic bias is widely recognised as an ethical consideration for recommendation models, sentiment analysis, and other AI marketing systems. Such bias, rooted in unrepresentative datasets and inadequate models, can be segmented into design, contextual, and application types (Akter et al., 2022). Design bias emerges because of data, model or method factors. Contextual bias is caused by cultural, social and personal components. Application bias originates from product, price, place and promotion dynamics. In recommendation modelling, customers could receive unfair outcomes in terms of value creation, value delivery (e.g. channels), and value management (e.g. price). In fact, Lambrecht & Tucker (2018) established that, because of embedded bias behind Facebook's advertisement placement algorithm, propositions of STEM professions on the platform were shown more frequently to males. Furthermore, overreliance on algorithmic recommendations can limit true choice and spontaneous discovery (Bitar & El Alam, 2024). Hence, commercial leaders be responsible for auditing, detecting and reducing potential systematic biases, to ensure fairness. Notably, AI is raising concerns about the privacy and manipulation of customer behaviour (Potwora et al., 2024), where the ramifications of a monetisable marketing data economy is being referred to as a form of 'surveillance capitalism' (Zuboff, 2023). Indeed, Europe's General Data Protection Regulation (GDPR) of May 2018 extends AI's legal and ethical debate. Li et al. (2020) demonstrate that the acceptance of AI in marketing increases with greater faith in the dependability and ethics of these systems. Hence, companies should uphold data governance frameworks, balance customisation with privacy, and remain transparent (Zhang et. al., 2018).

7 Conclusion

In 1950, an economist by the name of Joseph Schumpeter examined the competitive dynamics of manufacturing markets. Schumpeter used social science and game theory to establish connections between an organisation's rate of innovation and survival prospects. The research complemented the same scholar's prior definition of 'Creative Destruction' in 1942, a process whereby innovation displaces existing products, technologies, institutions, and economic structures (Investopedia, 2025). The phenomena, driven by entrepreneurs and the nature of competition, propels capitalism forward, whilst validating the Unique Selling Points (USPs) of generational pioneers. Furthermore, 'Diffusion of Innovations', a book by Everett Rogers published in 1962, describes the spread of new ideas in society, through 'innovators', 'early adopters', the 'early majority', and the 'late majority'. To some extent, AI marketing models and systems, for the purpose of sentiment analysis and recommendation engines, are forms innovation for academics, industry professionals and companies to deliberate. In conclusion, the research project involved developing Transformer-Based Sentiment Analysis Models (TBSAMs), Machine Learning Recommendation Models (MLRMs), and a hybrid Content-Based Recommendation System (CBRS), using Stanford Network Analysis Project (SNAP) Amazon reviews data for video games. For the former, Round 3 of the DistilBERT TBSAMs yielded superior performance metrics compared to Rounds 1 and 2. For the MLRMs, the Logistic Regression (LR) classifier outperformed every K-Nearest Neighbours (KNN) option and formed the probabilistic structure of the CBRS. Lastly, cosine similarity completed the item-specific elements of the CBRS. Overall, these methods are applicable to industries beyond entertainment, domains other than marketing, and objectives believed to be non-commercial.

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9 Appendix

Table A.3.1. SNAP & Amazon Reviews for Video Games – Variables (Sentiment)

Variable	Description	Example
productid	Amazon Standard Identification Number	B000068VC6
title	Full Amazon listing title of the product	Lara Croft Tomb Raider: The Angel of Darkness
price	Amazon listing price of the product	6.99
score	User rating of the product (1.0 - 5.0 stars)	3.0
time	Time of the user review (UNIX)	1217116800
summary	Text of the user review summary	not bad
review	Text of the user review	the game's good, probably better on a ps2 or Xbox, sometimes a little harder to control on the pc...
userreviewid	Unique user review identification for listed product	984UZX13L7GJ
categories	Categorical information of the listed product	Video Games, PC, Action, Role-Playing
description	Description of the listed product	follow the sultry and super-tough lara croft as she fights to clear her name in a game of blood and betrayal
console	Console of the listed product extracted from 'categories'	PC & Mac

Figure A.3.1. Score Distribution (Bar Chart)

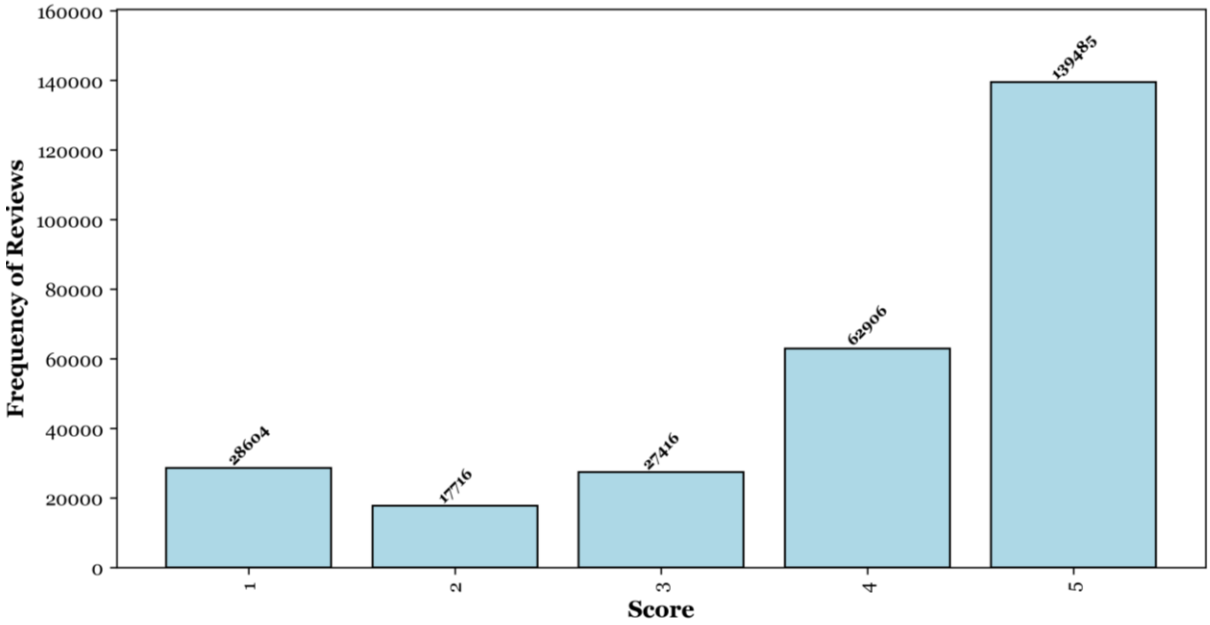

Figure B.3.1. Price Distribution (Box Plot) – Outliers (Panel 1) & No Outliers (Panel 2)

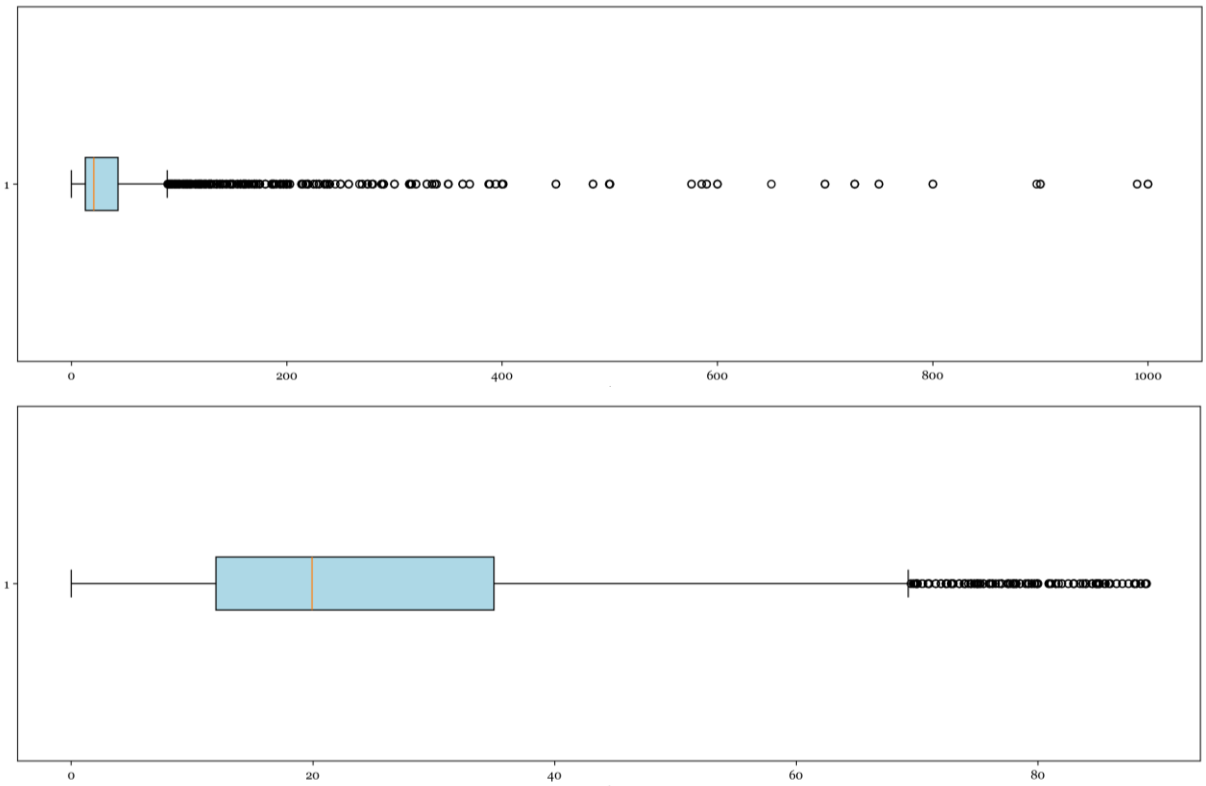

Figure C.3.1. Console Distribution (Bar Chart) – Outliers (Panel 1) & No Outliers (Panel 2)

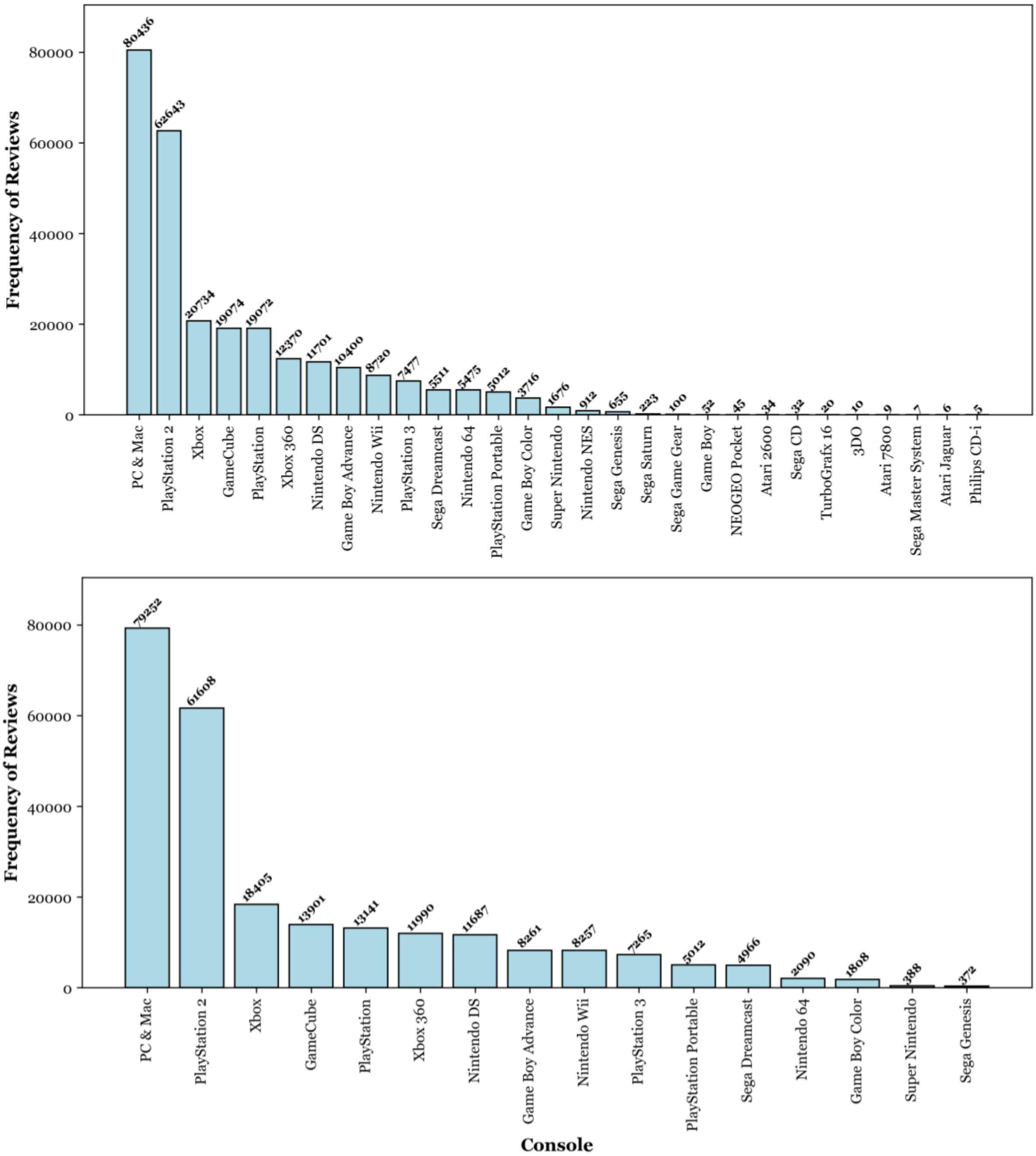

Table A.3.2. SNAP & Amazon Reviews for Video Games – Variables (Recommendation)

Variable	Description	Example
productid	Amazon Standard Identification Number	B00008URUA
title	Full Amazon listing title of the product	Final Fantasy X-2
description	Description of the listed product	join yuna in mission-based nonlinear gameplay as she and her companions travel...
price	Amazon listing price of the product	14.99
price_normalised	Normalised Amazon listing price of the product	-0.57851
console	Console of the listed product without encoded representation	PlayStation 2*
console_consolename	Dummy variable indicating presence (1) or non-presence (0) of console	1 – console_PlayStation 2 * 0 – console_PC & Mac 0 – console_Nintendo Wii 0 – console_Xbox ...
review_counts	Number of reviews of the listed product	62
average_score	Average user rating of the product	3.70978
adjusted_average_score (AAS)	Bayesian adjusted average user rating of the product	3.70978
adjusted_average_sentiment_score	Dummy variable of positive (1) or non-positive (0) sentiment from AAS	1
average_bert_pred_prob_pos	Average DistilBERT probability of review sentiment as positive	0.57116
adjusted_average_bert_pred_prob_pos	Adjusted average DistilBERT probability of review sentiment as positive	0.57116
average_sentiment_vader	Rule-based average VADER compound sentiment score	0.77320
adjusted_average_sentiment_vader	Rule-based adjusted average VADER compound sentiment score	0.77320

Table B.3.2. SNAP & Amazon Reviews for Video Games - Descriptive Statistics (Recommendation)

Variable	Mean	Standard Deviation	Minimum	1st Quartile	Median	3rd Quartile	Maximum
price	26.55443	18.71885	0.74000	12.61250	20.16500	36.47500	88.99000
price_normalised	0.04929	1.01619	-1.35209	-0.70757	-0.29757	0.58785	3.43873
review_counts	10.13404	13.00782	4.00000	5.00000	6.00000	10.00000	241.00000
average_score	3.93433	0.69889	1.25000	3.55556	4.00000	4.46961	5.00000
adjusted_average_score	3.53572	0.51964	1.60996	3.21429	3.42857	3.72727	5.00000
average_bert_pred_prob_pos	0.59575	0.13337	0.12680	0.51251	0.60532	0.68963	0.96450
adjusted_average_bert_pred_prob_pos	0.55529	0.08005	0.23905	0.50476	0.54772	0.59021	0.86634
average_bert_pred_prob_nonpos	0.40425	0.13337	0.03550	0.31037	0.39468	0.48749	0.87320
adjusted_average_bert_pred_prob_nonpos	0.44471	0.08005	0.13366	0.40979	0.45228	0.49524	0.76095
average_sentiment_vader	0.60999	0.28672	-0.78635	0.45900	0.66429	0.83635	0.99022
adjusted_average_sentiment_vader	0.56446	0.15493	-0.42259	0.48328	0.57835	0.64591	0.96850

Table C.3.2. SNAP & Amazon Reviews for Video Games – Variance Inflated Factor (VIF)

Variable	VIF (Adjusted)	VIF (Unadjusted)
price	3.09594	3.14530
average_score	–	71.06164
adjusted_average_score	154.23697	–
average_bert_pred_prob_pos	–	51.43485
adjusted_average_bert_pred_prob_pos	134.28386	–
average_bert_pred_prob_nonpos	–	6.23222
adjusted_average_bert_pred_prob_nonpos	11.76807	–
average_sentiment_vader	–	7.81199
adjusted_average_sentiment_vader	22.48030	–
review_counts	1.91507	1.62233

Figure A.3.2. DistilBERT Distribution (Violin Plot)
– Average (Panel 1) & Adjusted Average (Panel 2)

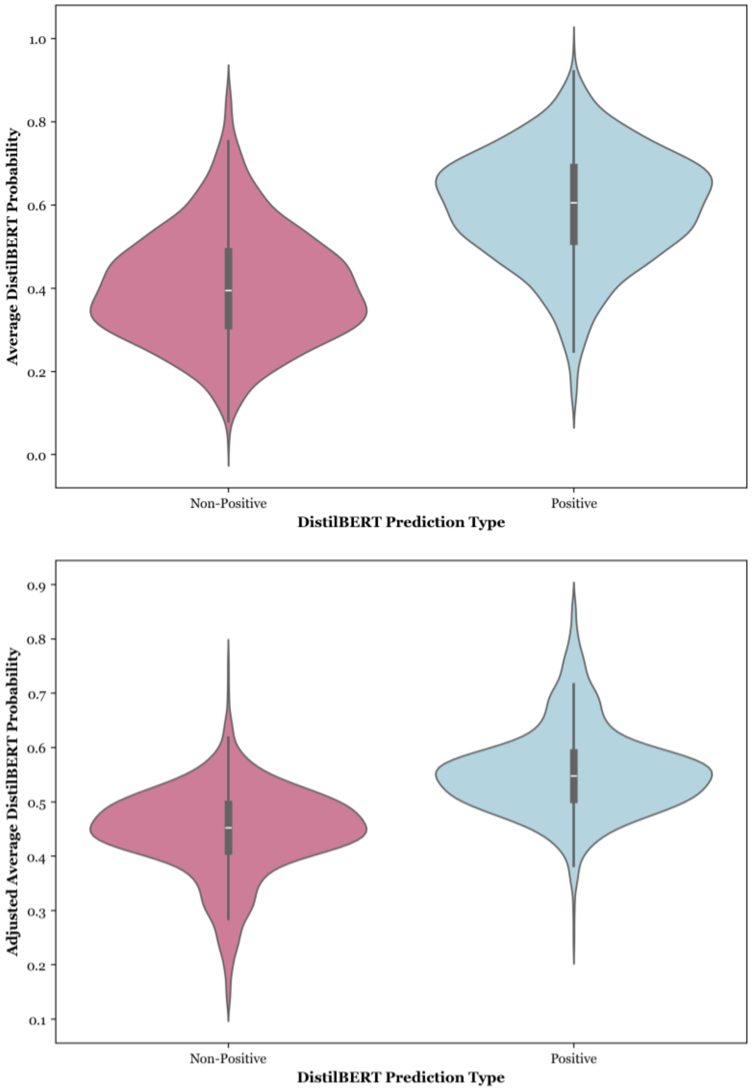

Figure B.3.2. DistilBERT Distribution (KDE Plot)
– Average (Panel 1) & Adjusted Average (Panel 2)

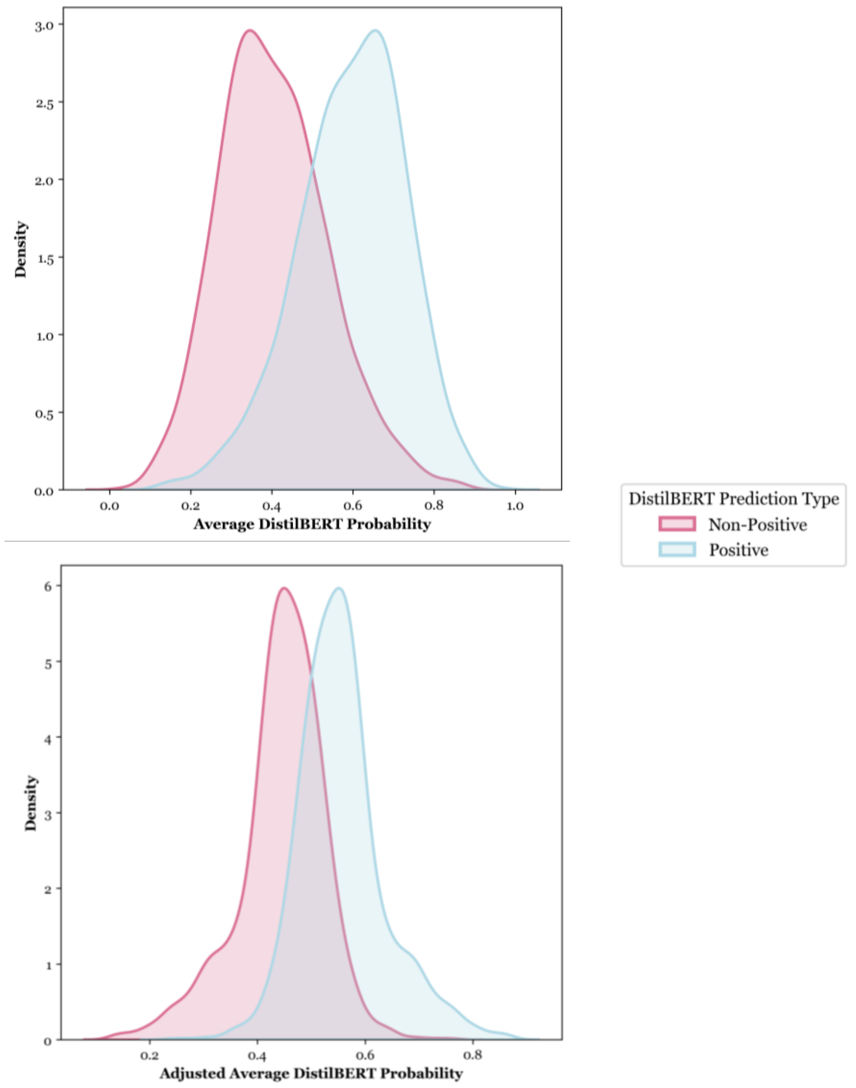

Figure C.3.2. Price & Review Counts Correlation

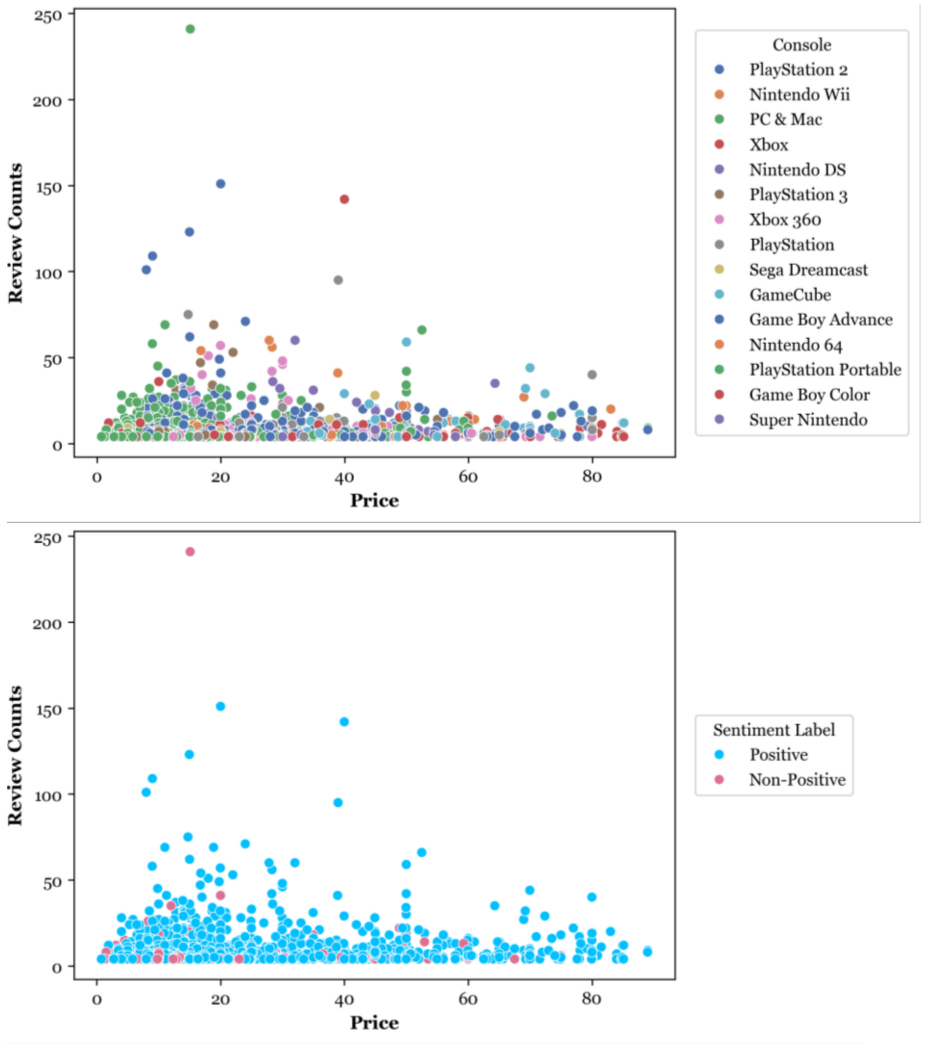

Figure D.3.2. Price & Adjusted Average Score Correlation

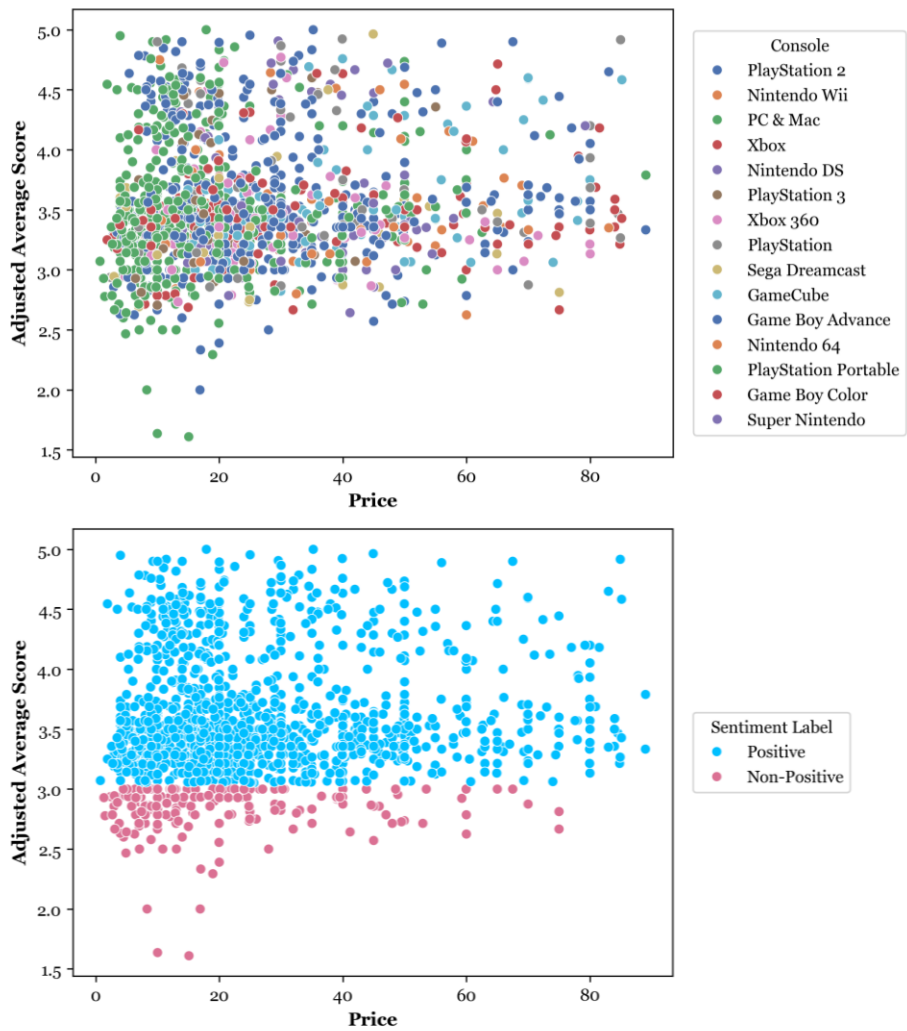

Figure E.3.2. Adjusted Average Score &
Review Counts Correlation

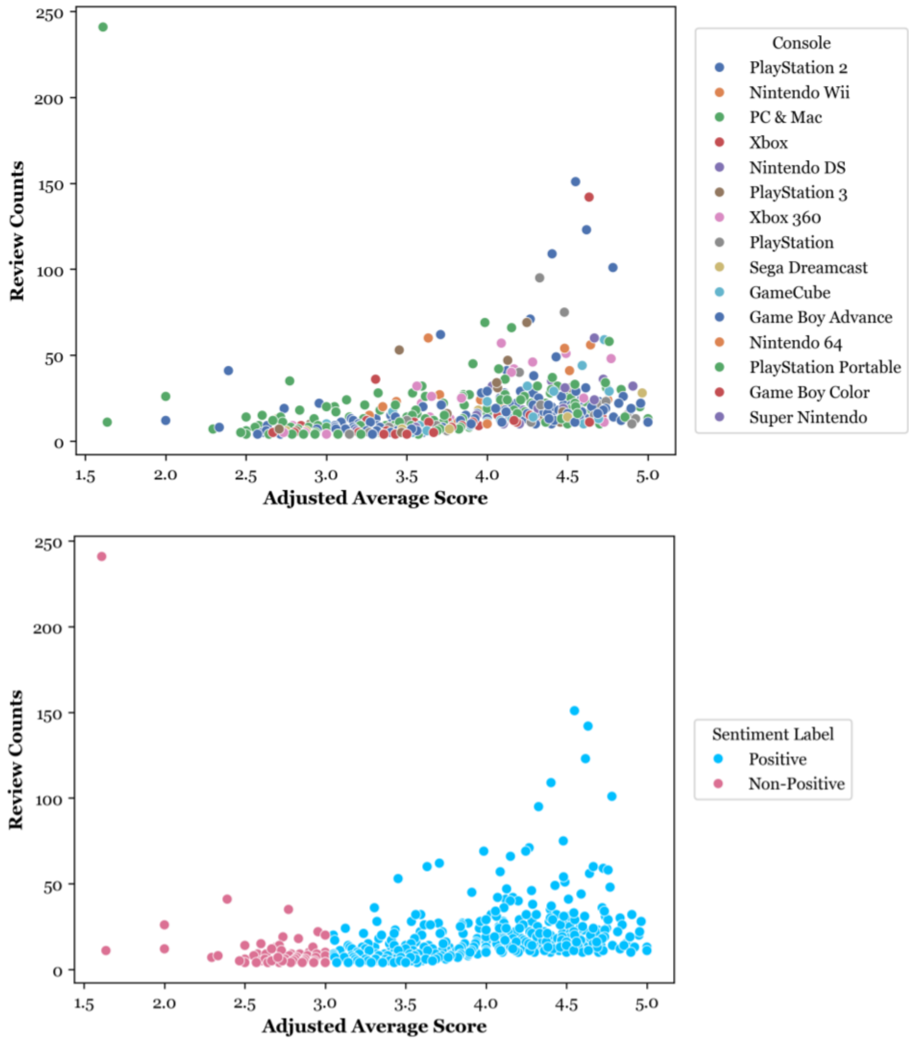

Figure F.3.2. Adjusted Average Score &
Adjusted Average VADER Sentiment Score Correlation

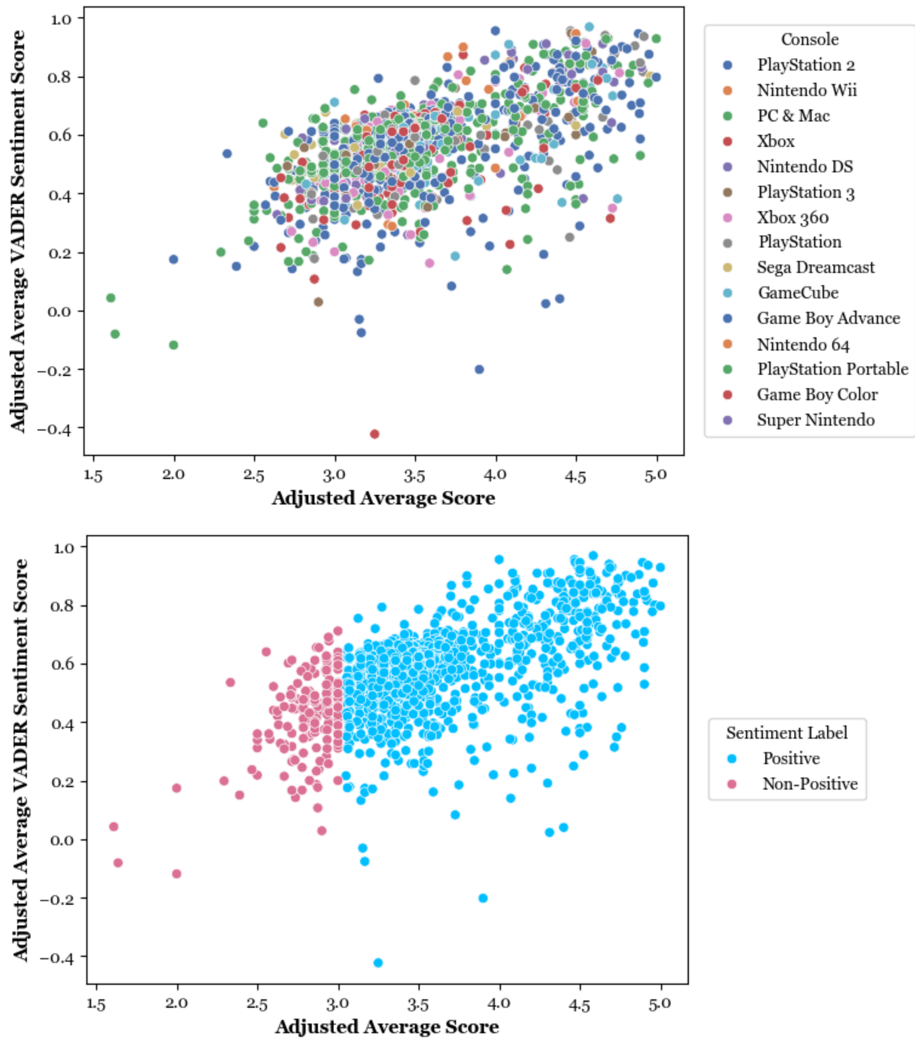

Figure G.3.2. Adjusted Average Score & Adjusted Average DistilBERT Probability Correlation

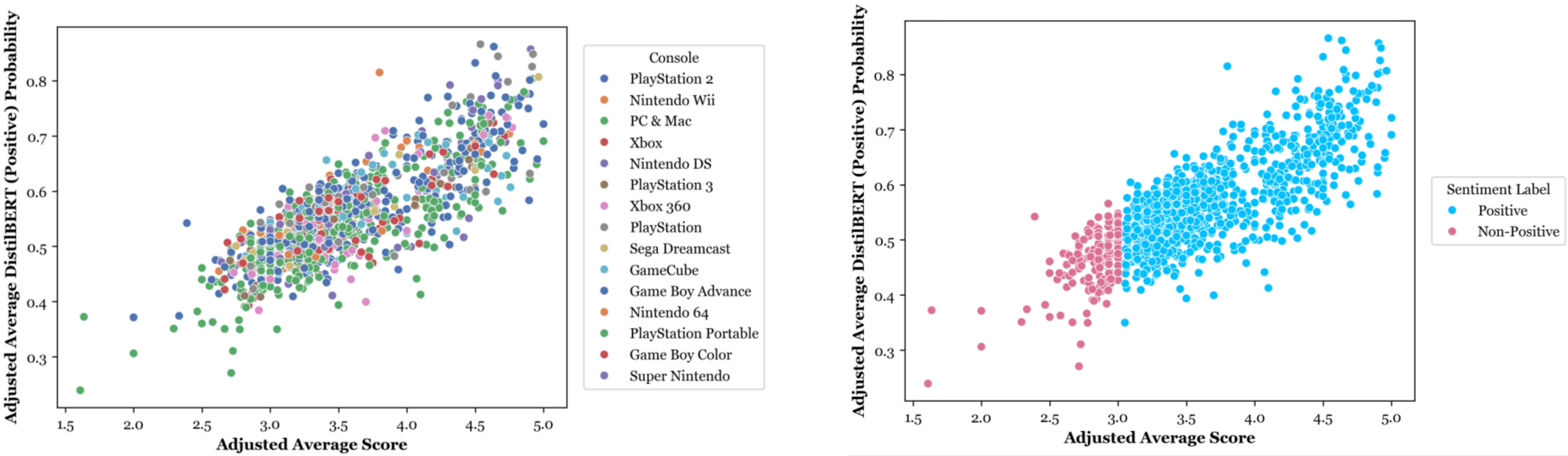

Table A.5.1. DistilBERT Transformer-Based Sentiment Analysis Models (TBSAMs) – Performance Report

Round 1	Classes & Averages	Support	Precision	Recall	F1-Score	Accuracy	Misclassification Rate	AUC
Training	Negative	1,754	0.49	0.67	0.57	0.70011	0.29989	0.8691
	Neutral	1,020	0.25	0.40	0.31			0.7453
	Positive	7,163	0.90	0.75	0.82			0.8526
	Macro Average	–	0.55	0.60	0.56			–
	Weighted Average	–	0.76	0.70	0.72			–
Validation	Negative	219	0.46	0.67	0.55	0.69082	0.30918	0.8590
	Neutral	128	0.28	0.43	0.34			0.7470
	Positive	895	0.90	0.73	0.81			0.8478
	Macro Average	–	0.55	0.61	0.56			–
	Weighted Average	–	0.76	0.69	0.71			–
Test	Negative	219	0.46	0.59	0.52	0.67900	0.32100	0.8645
	Neutral	128	0.18	0.29	0.22			0.6798
	Positive	896	0.90	0.76	0.82			0.8468
	Macro Average	–	0.51	0.54	0.52			–
	Weighted Average	–	0.75	0.68	0.71			–
Round 2								
Training	Negative	3,508	0.47	0.71	0.56	0.68616	0.31384	0.8617
	Neutral	2,039	0.24	0.34	0.28			0.7295
	Positive	14,326	0.90	0.73	0.81			0.8405
	Macro Average	–	0.53	0.59	0.55			–
	Weighted Average	–	0.76	0.69	0.71			–
Validation	Negative	438	0.45	0.69	0.54	0.68317	0.31683	0.8560
	Neutral	255	0.26	0.35	0.30			0.7141
	Positive	1,791	0.90	0.73	0.80			0.8377
	Macro Average	–	0.53	0.59	0.55			–
	Weighted Average	–	0.75	0.68	0.71			–
Test	Negative	439	0.47	0.74	0.57	0.69296	0.30704	0.8649
	Neutral	255	0.22	0.28	0.25			0.7212
	Positive	1,791	0.90	0.74	0.81			0.8470
	Macro Average	–	0.53	0.59	0.54			–

Round 3	Classes & Averages	Support	Precision	Recall	F1-Score	Accuracy	Misclassification Rate	AUC
Training	Non-Positive	5,547	0.57	0.75	0.65	0.77240	0.22760	–
	Positive	14,326	0.89	0.78	0.83			0.8466
	Macro Average	–	0.73	0.76	0.74			–
	Weighted Average	–	0.80	0.77	0.78			–
Validation	Non-Positive	693	0.59	0.77	0.67	0.78583	0.21417	–
	Positive	1,791	0.90	0.79	0.84			0.8599
	Macro Average	–	0.74	0.78	0.76			–
	Weighted Average	–	0.81	0.79	0.79			–
Test	Non-Positive	694	0.56	0.76	0.65	0.76901	0.23099	–
	Positive	1,791	0.89	0.77	0.83			0.8474
	Macro Average	–	0.73	0.77	0.74			–
	Weighted Average	–	0.80	0.77	0.78			–

Table B.5.1. DistilBERT TBSAMs – Training & Validation Loss

DistilBERT Round	Loss Type	Epoch 1	Epoch 2	Epoch 3	Epoch 4	Epoch 5
Round 1	Training Loss	1.073	1.019	0.966	–	–
	Validation Loss	1.041	0.985	0.938	–	–
Round 2	Training Loss	1.058	0.985	0.932	0.898	0.880
	Validation Loss	1.021	0.957	0.922	0.889	0.872
Round 3	Training Loss	0.646	0.565	0.528	0.509	0.502
	Validation Loss	0.589	0.524	0.496	0.508	0.472

Figure A.5.1. DistilBERT TBSAMs – Receiver Operating Characteristic (ROC) Curves per Class (Round 1)

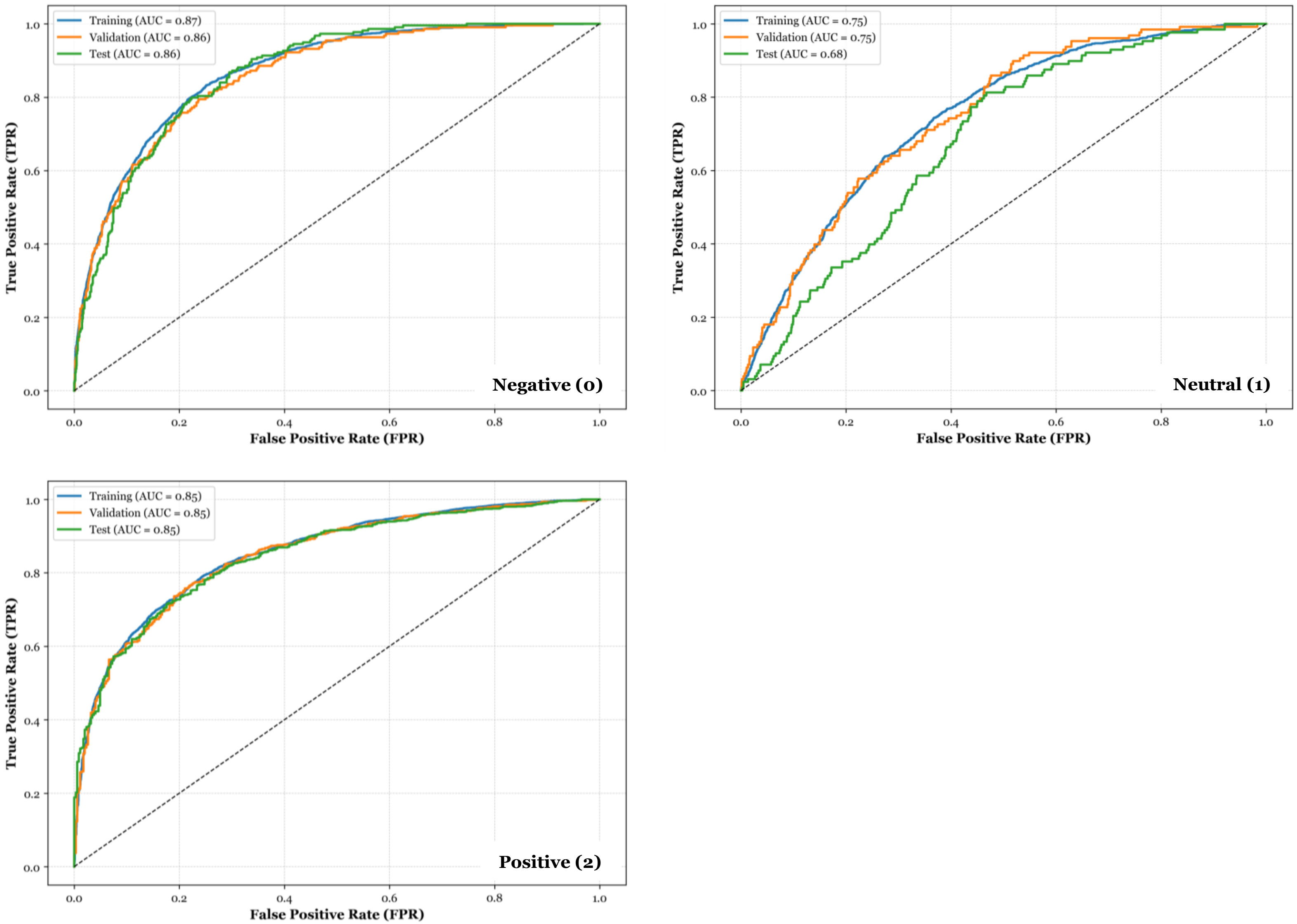

Figure B.5.1. DistilBERT TBSAMs – Receiver Operating Characteristic (ROC) Curves per Class (Round 2)

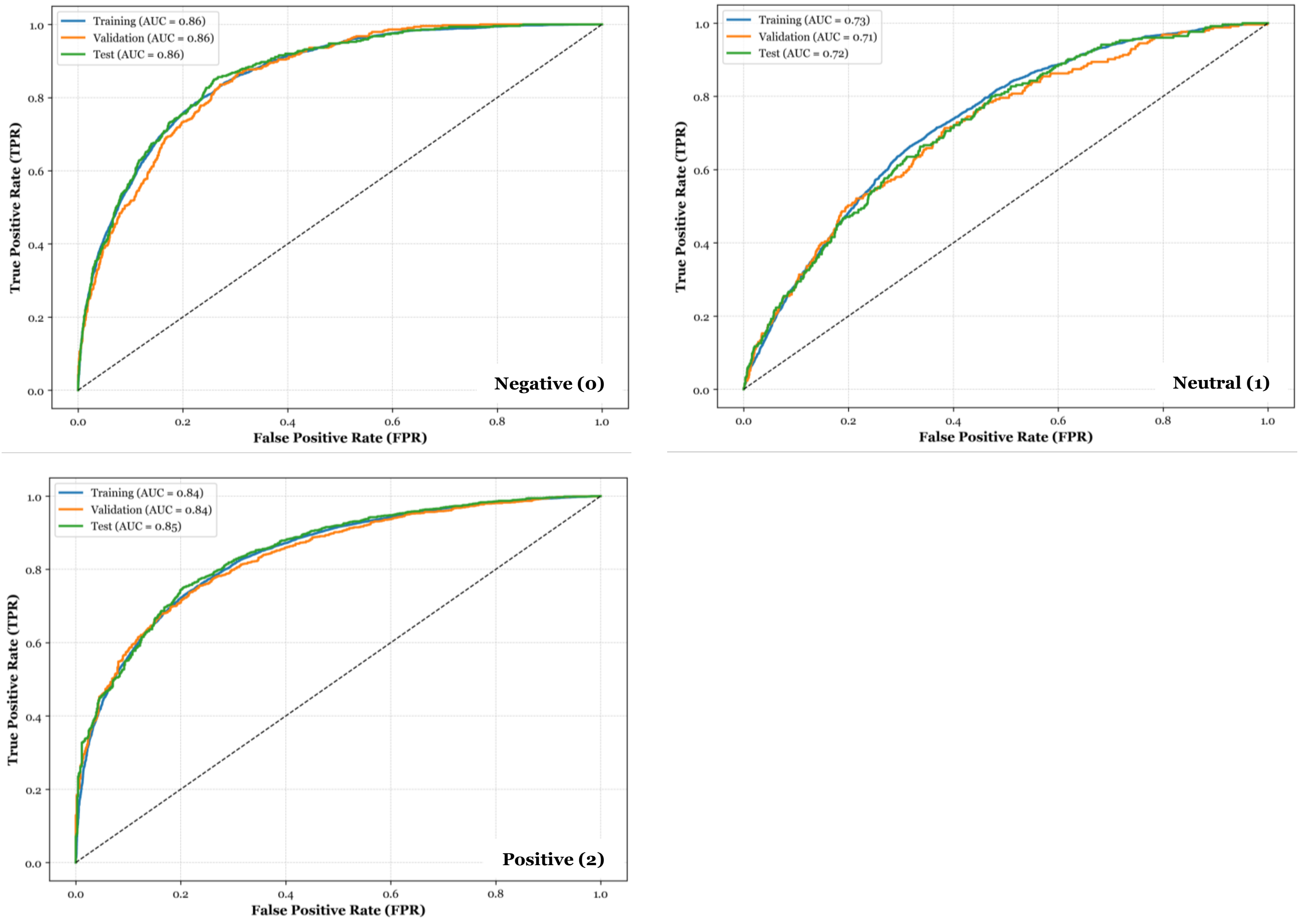

Figure C.5.1. DistilBERT TBSAMs – Receiver Operating Characteristic (ROC) Curves for Positive Class (Round 3)

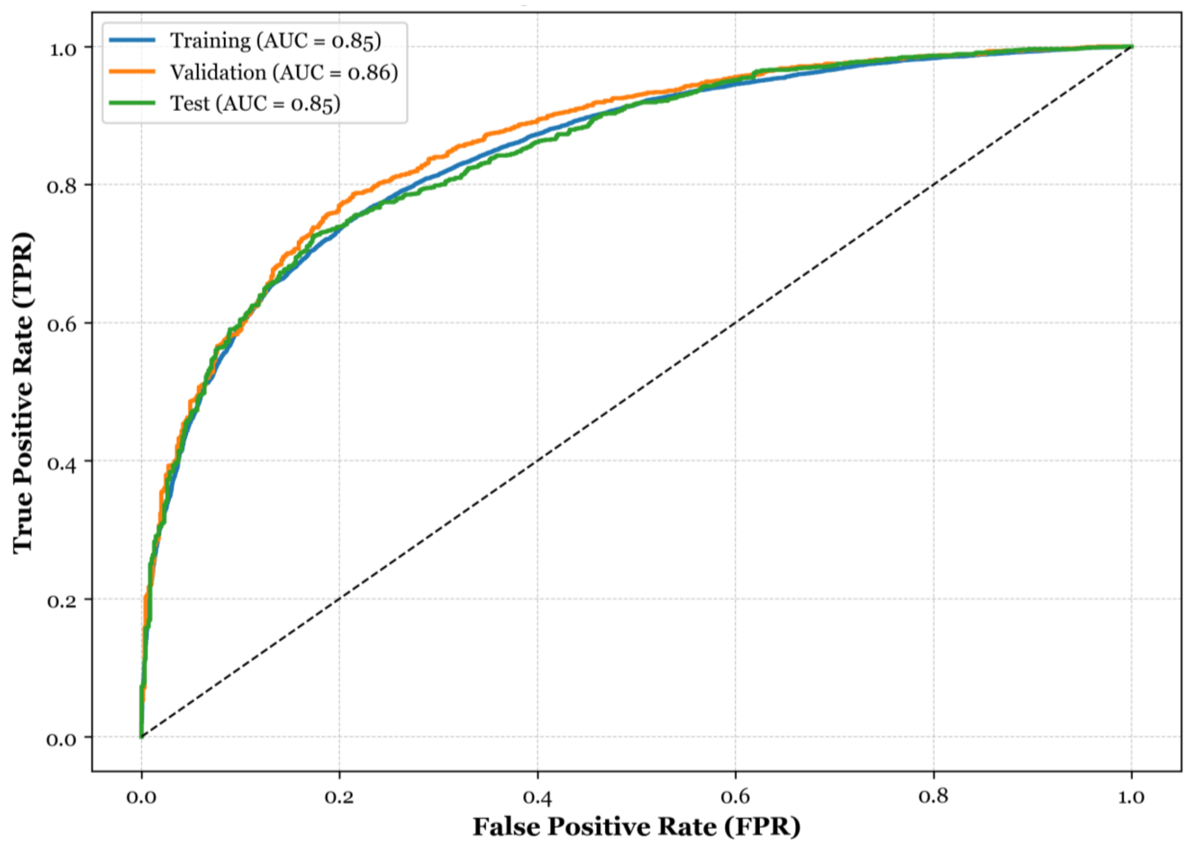

Figure D.5.1. DistilBERT TBSAMs – Training & Validation Loss (Round 3)

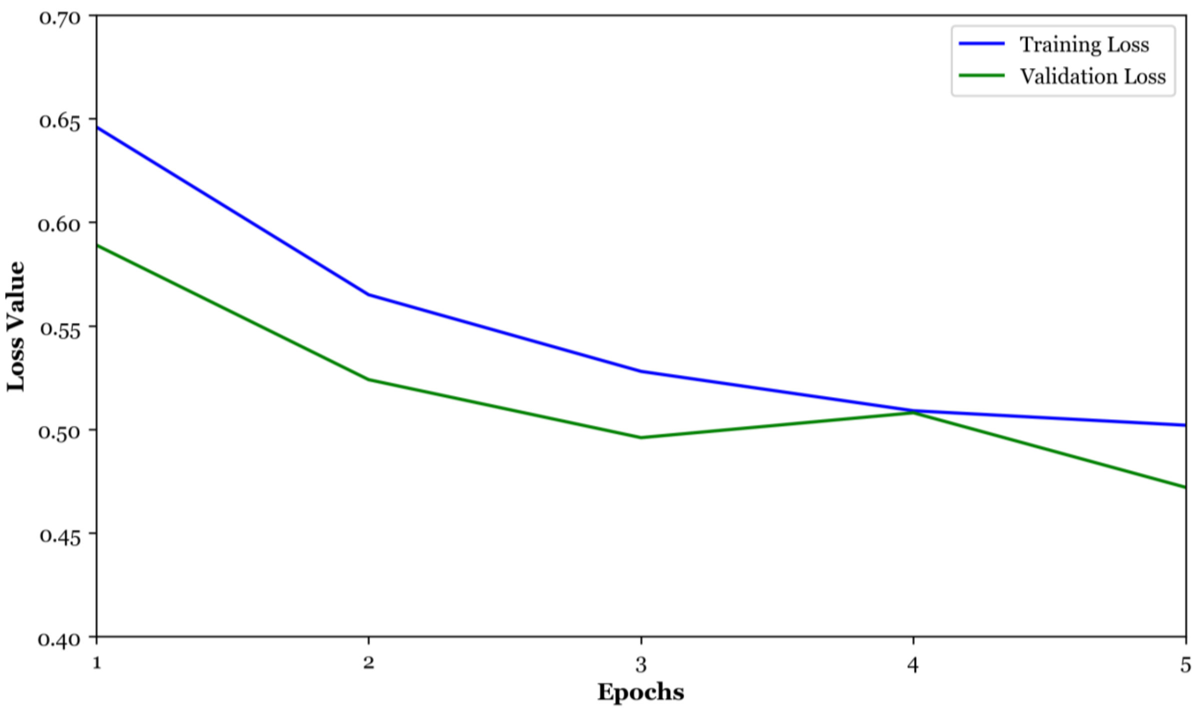

Figure E.5.1. DistilBERT TBSAMs – DistilBERT Distribution (Violin Plot) (Round 3)

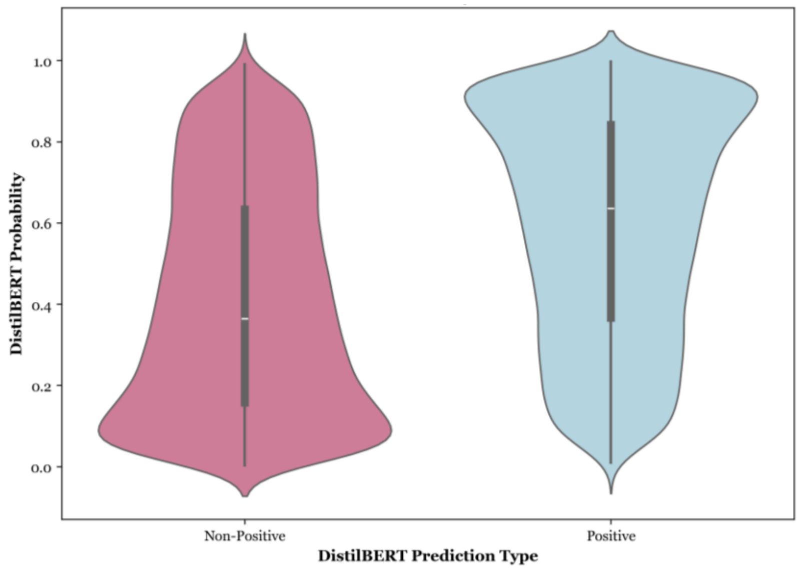

Figure F.5.1. DistilBERT TBSAMs – DistilBERT Distribution (Kernel Density Estimation Plot) (Round 3)

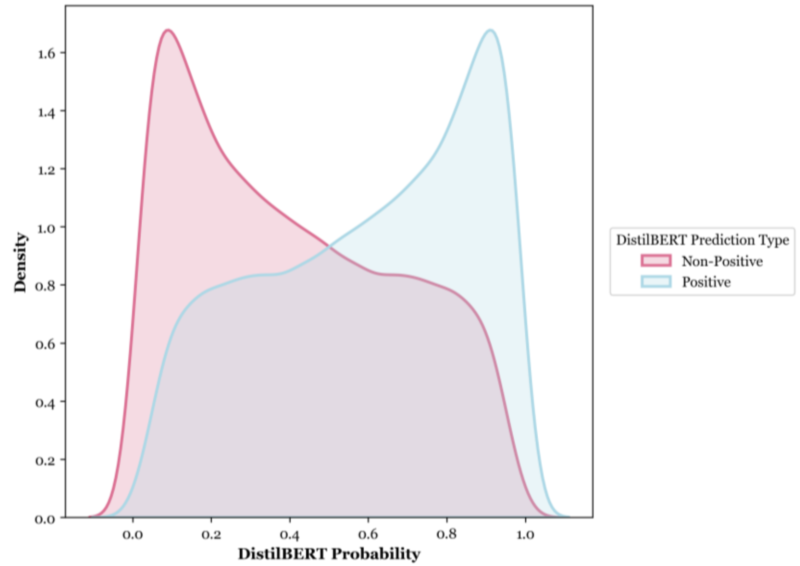

Figure G.5.1. DistilBERT TBSAMs – Sentiment Distribution (Bar Chart) (Round 3)

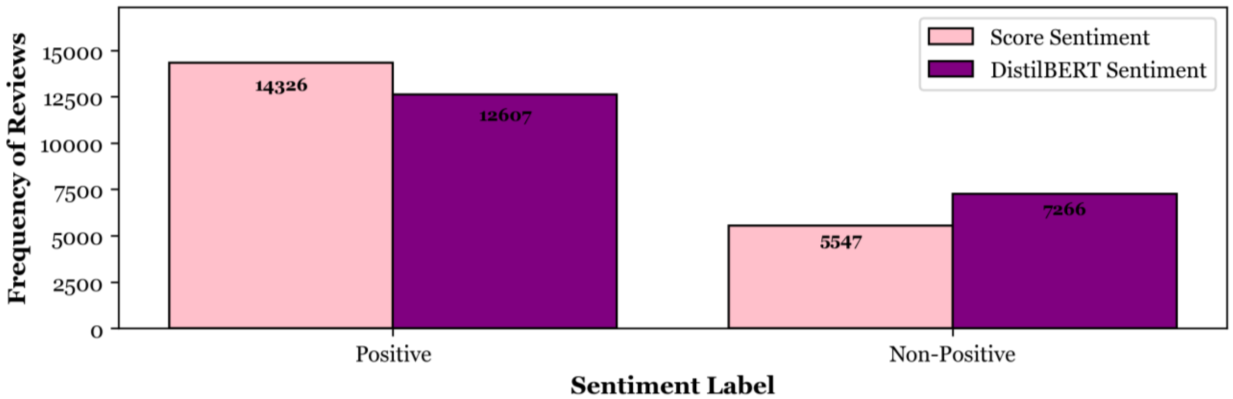

Table A.5.2. Logistic Regression (LR) Machine Learning Recommendation Model (MLRM) – Performance Report

Phase	Classes & Averages	Support	Precision	Recall	F1-Score	Accuracy	Misclassification Rate	AUC
Training	**Not Recommend (NP)**	143	0.81	0.98	0.89	0.96900	0.03100	–
	Recommend (P)	985	1.00	0.97	0.98			0.9965
	Macro Average	–	0.91	0.97	0.94			–
	Weighted Average	–	0.97	0.97	0.97			–
Test	**Not Recommend (NP)**	36	0.50	0.39	0.44	0.87230	0.12770	–
	Recommend (P)	246	0.91	0.94	0.93			0.8498
	Macro Average	–	0.71	0.67	0.68			–
	Weighted Average	–	0.86	0.87	0.87			–

Table B.5.3. Logistic Regression (LR) Machine Learning Recommendation Model (MLRM) – *description* Coefficients (Sample)

Word	Coefficient (Log-Odds)	Logistic Equation	Odds Ratio	Odds Change (%)
devastating	- 1.01281	$e^{-1.01281}$	0.36320	- 63.680
midnight	- 0.96708	$e^{-0.96708}$	0.38019	- 61.981
unlimited	- 0.86330	$e^{-0.86330}$	0.42177	- 57.823
atlantis	- 0.79016	$e^{-0.79016}$	0.45377	- 54.623
ultimate	- 0.77141	$e^{-0.77141}$	0.46236	- 53.764
historical	- 0.74546	$e^{-0.74546}$	0.47452	- 52.548
dragonflies	- 0.73539	$e^{-0.73539}$	0.47932	- 52.068
alchemist	- 0.70167	$e^{-0.70167}$	0.49576	- 50.424
matrix	- 0.61617	$e^{-0.61617}$	0.54001	- 45.999
players	+ 0.75452	$e^{+0.75452}$	2.12659	+ 112.659
story	+ 0.91445	$e^{+0.91445}$	2.49540	+ 149.540
monsters	+ 0.65852	$e^{+0.65852}$	1.93193	+ 93.193
characters	+ 0.55074	$e^{+0.55074}$	1.73454	+ 73.454
everquest	+ 0.46648	$e^{+0.46648}$	1.59437	+ 59.437
campaigns	+ 0.45307	$e^{+0.45307}$	1.57313	+ 57.313
mystery	+ 0.44565	$e^{+0.44565}$	1.56150	+ 56.150
restaurant	+ 0.42457	$e^{+0.42457}$	1.52893	+ 52.893
control	+ 0.31028	$e^{+0.31028}$	1.36381	+ 36.381

Table C.5.2. Logistic Regression (LR) Machine Learning Recommendation Model (MLRM) – Model Coefficients

Variable	Coefficient (Log-Odds)	Logistic Equation	Odds Ratio	Odds Change (%)
price_normalised	+ 0.29597	$e^{+\,0.29597}$	1.34443	+ 34.443
adjusted_average_bert_pred_prob_pos	+ 11.91229	$e^{+\,11.91229}$	149087.6997	+ 14908669.965
console_Game Boy Advance	+ 0.32383	$e^{+\,0.32383}$	1.38241	+ 38.241
console_Game Boy Color	- 0.52945	$e^{-\,0.52945}$	0.58893	- 41.107
console_GameCube	+ 1.22016	$e^{+\,1.22016}$	3.38773	+ 238.773
console_Nintendo 64	- 0.34067	$e^{-\,0.34067}$	0.71138	- 28.862
console_Nintendo DS	+ 0.14285	$e^{+\,0.14285}$	1.15359	+ 15.359
console_Nintendo Wii	- 0.25803	$e^{-\,0.25803}$	0.77258	- 22.742
console_PC & Mac	- 0.19576	$e^{-\,0.19576}$	0.82223	- 17.777
console_PlayStation	- 0.26320	$e^{-\,0.26320}$	0.76848	- 23.152
console_PlayStation 2	+ 0.10240	$e^{+\,0.10240}$	1.10782	+ 10.782
console_PlayStation 3	+ 0.05419	$e^{+\,0.05419}$	1.05567	+ 5.567
console_PlayStation Portable	+ 0.70276	$e^{+\,0.70276}$	2.01927	+ 101.927
console_Sega Dreamcast	- 0.84526	$e^{-\,0.84526}$	0.42958	- 57.042
console_Super Nintendo	+ 0.97664	$e^{+\,0.97664}$	2.65611	+ 165.611
console_Xbox	+ 0.71913	$e^{+\,0.71913}$	2.05285	+ 105.285
console_Xbox 360	+ 0.03442	$e^{+\,0.03442}$	1.03502	+ 3.502

Figure A.5.2. LR MLRM – Receiver Operating Characteristic (ROC) Curve

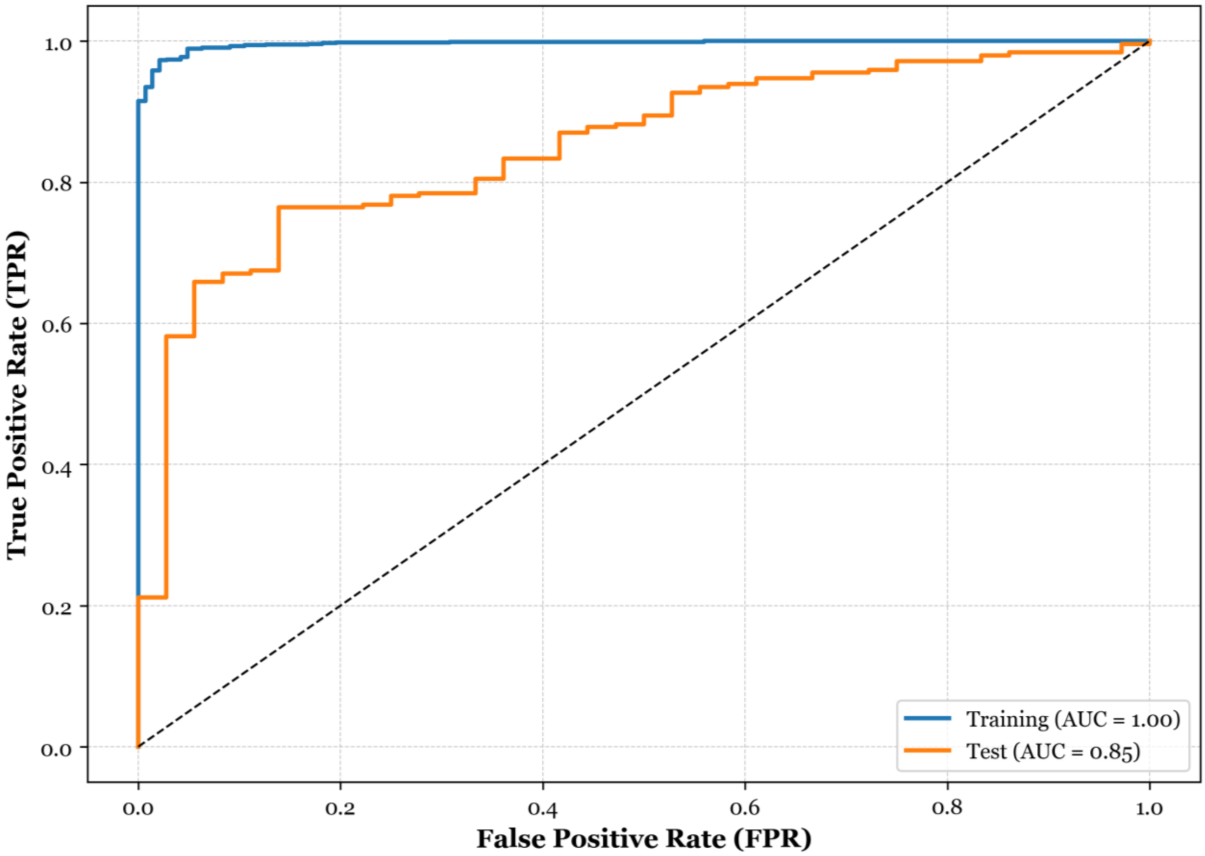

Table D.5.2. K-Nearest Neighbours (KNN) & Manhattan Distance MLRM – Performance Report (Test)

Neighbours	Classes & Averages	Support	Precision	Recall	F1-Score	Accuracy	Misclassification Rate	AUC
K = 1	**Not Recommend (NP)**	36	0.19	0.14	0.16	0.81560	0.18440	–
	Recommend (P)	246	0.88	0.91	0.90			0.5268
	Macro Average	–	0.54	0.53	0.53			–
	Weighted Average	–	0.79	0.82	0.80			–
K = 2	**Not Recommend (NP)**	36	0.15	0.19	0.17	0.75530	0.24470	–
	Recommend (P)	246	0.88	0.84	0.86			0.5197
	Macro Average	–	0.51	0.52	0.51			–
	Weighted Average	–	0.78	0.76	0.77			–
K = 3	**Not Recommend (NP)**	36	0.12	0.14	0.13	0.76240	0.23760	–
	Recommend (P)	246	0.87	0.85	0.86			0.5343
	Macro Average	–	0.50	0.50	0.50			–
	Weighted Average	–	0.78	0.76	0.77			–
K = 4	**Not Recommend (NP)**	36	0.18	0.67	0.29	0.58160	0.41840	–
	Recommend (P)	246	0.92	0.57	0.70			0.5967
	Macro Average	–	0.55	0.62	0.50			–
	Weighted Average	–	0.83	0.58	0.65			–
K = 5	**Not Recommend (NP)**	36	0.21	0.56	0.31	0.67730	0.32270	–
	Recommend (P)	246	0.91	0.70	0.79			0.6168
	Macro Average	–	0.56	0.63	0.55			–
	Weighted Average	–	0.82	0.68	0.73			–
K = 6	**Not Recommend (NP)**	36	0.15	0.78	0.26	0.42200	0.57800	–
	Recommend (P)	246	0.92	0.37	0.53			0.5989
	Macro Average	–	0.54	0.57	0.39			–
	Weighted Average	–	0.82	0.42	0.49			–

K = 7	**Not Recommend (NP)**	36	0.18	0.75	0.29	0.52840	0.47160	–
	Recommend (P)	246	0.93	0.50	0.65			0.6112
	Macro Average	–	0.56	0.62	0.47			–
	Weighted Average	–	0.84	0.53	0.60			–
K = 8	**Not Recommend (NP)**	36	0.13	0.94	0.23	0.17730	0.82270	–
	Recommend (P)	246	0.89	0.07	0.12			0.6051
	Macro Average	–	0.51	0.50	0.17			–
	Weighted Average	–	0.79	0.18	0.13			–

Table E.5.2. K-Nearest Neighbours (KNN) & Euclidean Distance MLRM – Performance Report (Test)

Neighbours	Classes & Averages	Support	Precision	Recall	F1-Score	Accuracy	Misclassification Rate	AUC
K = 1	**Not Recommend (NP)**	36	0.14	0.86	0.23	0.28370	0.71630	–
	Recommend (P)	246	0.91	0.20	0.33			0.5301
	Macro Average	–	0.52	0.53	0.28			–
	Weighted Average	–	0.81	0.28	0.31			–
K = 2	**Not Recommend (NP)**	36	0.12	0.92	0.22	0.15600	0.84400	–
	Recommend (P)	246	0.79	0.04	0.08			0.5115
	Macro Average	–	0.45	0.48	0.15			–
	Weighted Average	–	0.70	0.16	0.10			–
K = 3	**Not Recommend (NP)**	36	0.13	0.92	0.22	0.18440	0.81560	–
	Recommend (P)	246	0.86	0.08	0.14			0.5186
	Macro Average	–	0.50	0.50	0.18			–
	Weighted Average	–	0.77	0.18	0.15			–
K = 4	**Not Recommend (NP)**	36	0.12	0.94	0.22	0.13480	0.86520	–
	Recommend (P)	246	0.67	0.02	0.03			0.5671
	Macro Average	–	0.39	0.48	0.12			–
	Weighted Average	–	0.60	0.13	0.06			–
K = 5	**Not Recommend (NP)**	36	0.13	0.94	0.22	0.15600	0.84400	–
	Recommend (P)	246	0.83	0.04	0.08			0.5475
	Macro Average	–	0.48	0.49	0.15			–
	Weighted Average	–	0.74	0.16	0.10			–
K = 6	**Not Recommend (NP)**	36	0.12	0.94	0.22	0.13480	0.86520	–
	Recommend (P)	246	0.67	0.02	0.03			0.5157
	Macro Average	–	0.39	0.48	0.12			–
	Weighted Average	–	0.60	0.13	0.06			–

K = 7	**Not Recommend (NP)**	36	0.12	0.94	0.22	0.13830	0.86170	–
	Recommend (P)	246	0.71	0.02	0.04			0.5222
	Macro Average	–	0.42	0.48	0.13			–
	Weighted Average	–	0.64	0.14	0.06			–
K = 8	**Not Recommend (NP)**	36	0.12	0.94	0.22	0.12770	0.87230	–
	Recommend (P)	246	0.50	0.01	0.02			0.5069
	Macro Average	–	0.31	0.48	0.12			–
	Weighted Average	–	0.45	0.13	0.04			–

Figure B.5.2. KNN MLRM – Optimisation of K Value

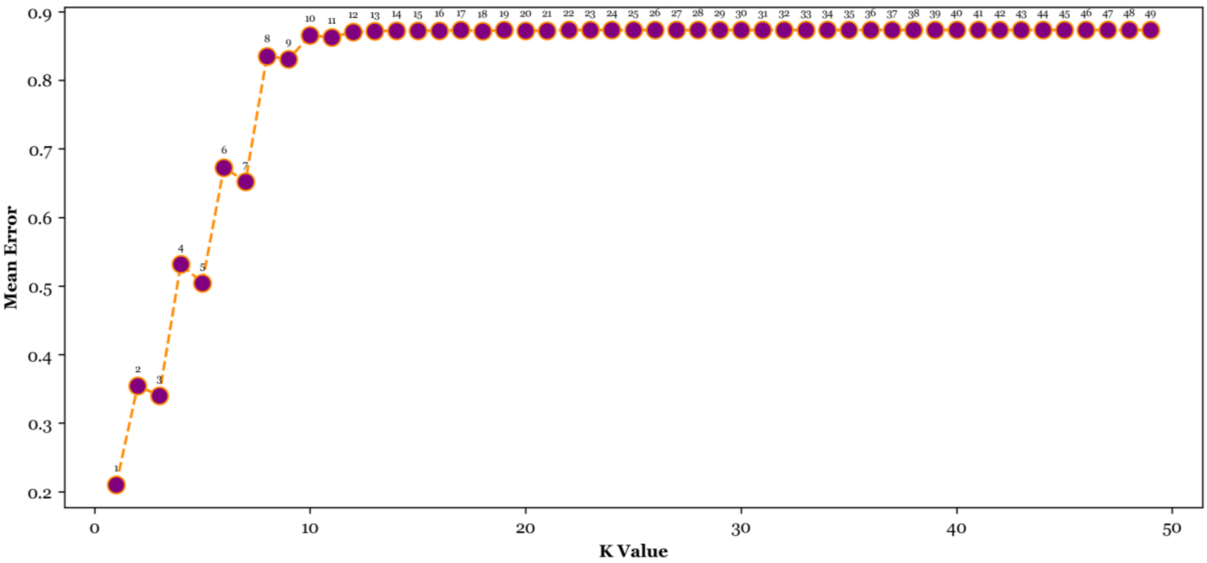

Figure C.5.2. KNN & Manhattan Distance MLRM –
Receiver Operating Characteristic (ROC) Curves

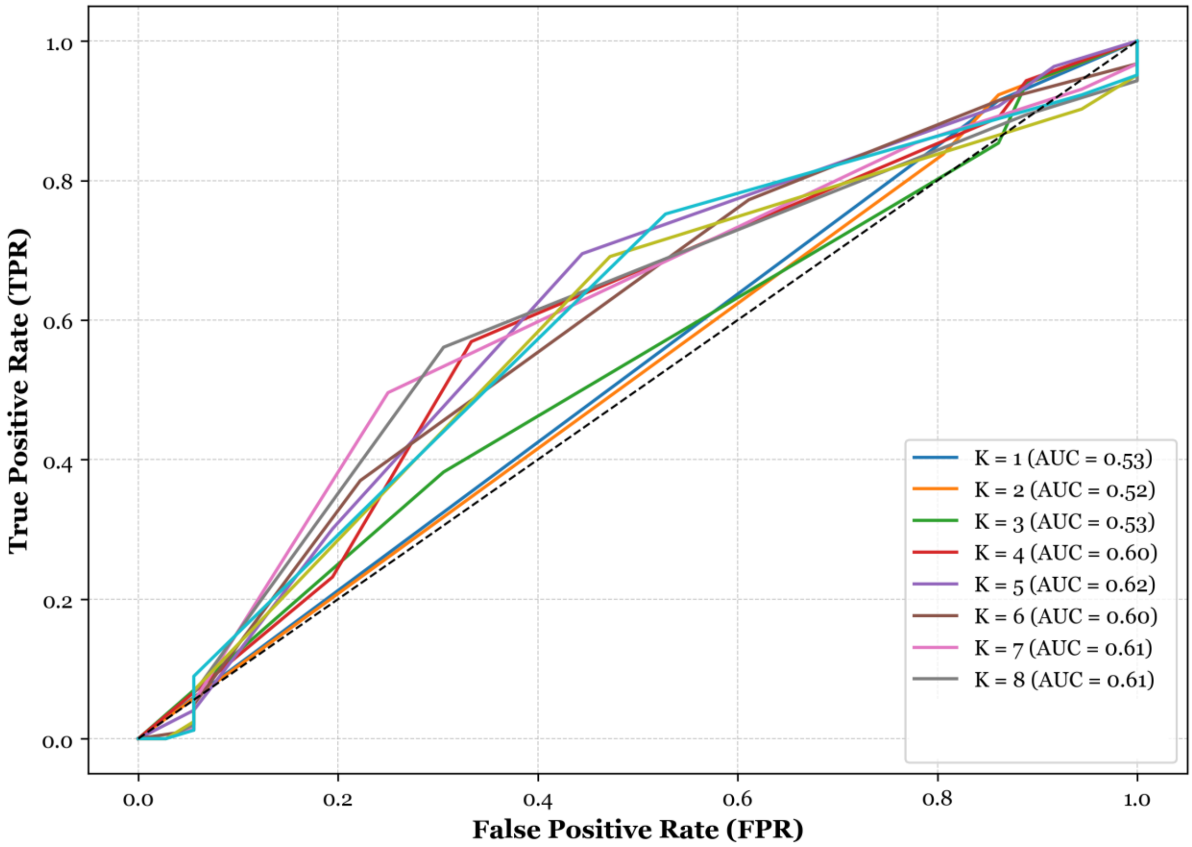

Figure D.5.2. KNN & Euclidean Distance MLRM –
Receiver Operating Characteristic (ROC) Curves

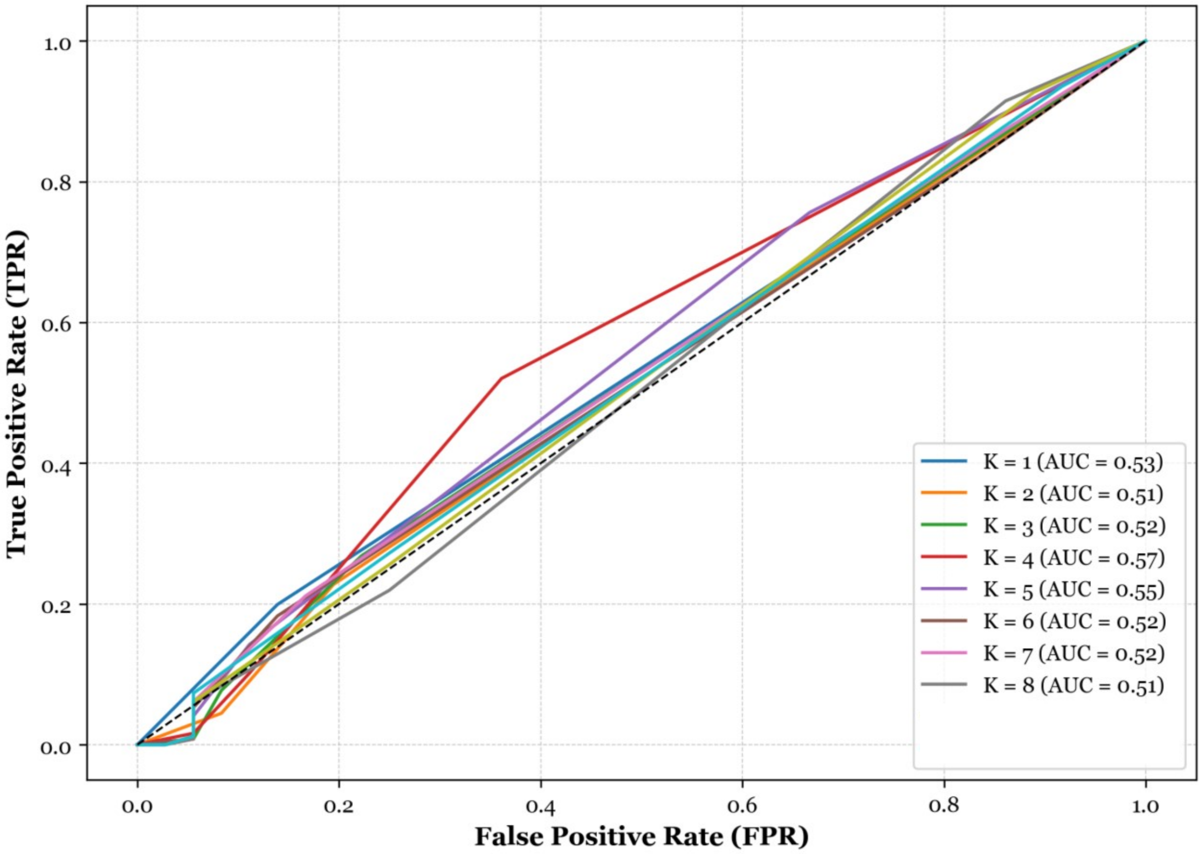

Table A.5.3. Content-Based Recommendation System (CBRS) – Highest & Lowest Logistic Probabilities of Positive Recommendation

Title	Console	Price	Logistic Probability
Highest			
Mario Kart: Super Circuit	Game Boy Advance	82.99	0.997253
Lego Star Wars II: The Original Trilogy	PlayStation 2	29.95	0.994129
Mario Party 7	GameCube	85.11	0.993439
Mario Kart DS	Nintendo DS	29.54	0.993222
Halo: Combat Evolved	Xbox	39.98	0.993157
CTR: Crash Team Racing	PlayStation	49.97	0.993145
Star Fox Assault	GameCube	69.95	0.991363
New Super Mario Bros	Nintendo DS	32.00	0.990830
Nintendogs Lab & Friends	Nintendo DS	41.94	0.990707
Wario World	GameCube	79.95	0.990386
Lowest			
Resident Evil 4	PC & Mac	11.49	0.550564
Dark Messiah of Might & Magic Elements	Xbox 360	19.45	0.551009
Left 4 Dead	PC & Mac	21.99	0.551493
Majestic Chess	PC & Mac	19.95	0.551664
High Heat Baseball 2003	PC & Mac	37.44	0.551730
Beyond Atlantis	PC & Mac	15.94	0.552502
Dragonshard	PC & Mac	3.99	0.552548
Space Channel 5	Sega Dreamcast	7.99	0.552844
RPG Maker II	PlayStation 2	39.98	0.553188
Icewind Dale 3 in 1 Boxset	PC & Mac	15.97	0.555162

Table B.5.3. Content-Based Recommendation System (CBRS) – Highest Probabilities of Positive Recommendation (per Console)

Title	Console	Price	Logistic Probability
Xbox 360			
Devil May Cry 4	Xbox 360	20.75	0.977325
Call of Duty 4: Modern Warfare	Xbox 360	29.98	0.975201
The Orange Box	Xbox 360	30.90	0.962334
Star Wars: The Force Unleashed	Xbox 360	15.96	0.959260
Burnout Paradise	Xbox 360	23.49	0.956976
Tom Clancy's Rainbow Six Vegas	Xbox 360	19.99	0.955374
Ninety-Nine Nights	Xbox 360	34.99	0.941515
Call of Duty 2	Xbox 360	20.00	0.940379
Lego Star Wars II: The Original Trilogy	Xbox 360	79.94	0.939453
Bioshock	Xbox 360	28.27	0.938299
PlayStation 2			
Lego Star Wars II: The Original Trilogy	PlayStation 2	29.95	0.994129
NBA Street	PlayStation 2	18.82	0.989516
Sly 2: Band of Thieves	PlayStation 2	13.99	0.984633
Ico	PlayStation 2	55.98	0.984523
PlayStation 2 Eye Toy	PlayStation 2	42.97	0.982069
WWE Smackdown! Shut Your Mouth	PlayStation 2	52.00	0.980211
Tony Hawk's Pro Skater 3	PlayStation 2	16.98	0.979038
James Bond 007: Everything or Nothing	PlayStation 2	28.98	0.977579
Star Wars Battlefront	PlayStation 2	44.98	0.976998
James Bond 007: Nightfire	PlayStation 2	29.97	0.976101
PC & Mac			
Age of Empires II, Gold Edition	PC & Mac	11.40	0.986904
The Sims Unleashed Expansion Pack	PC & Mac	19.95	0.973214
Diablo Battlechest [new version]	PC & Mac	20.80	0.972631
StarCraft	PC & Mac	18.49	0.966761
Star Wars Jedi Knight: Jedi Academy	PC & Mac	16.32	0.963257
Warcraft III Battle Chest	PC & Mac	19.99	0.961657
Halo: Combat Evolved	PC & Mac	18.60	0.959775
Pharaoh	PC & Mac	22.50	0.959209
WarCraft III Expansion: The Frozen Throne	PC & Mac	19.99	0.958125
Roller Coaster Tycoon	PC & Mac	8.95	0.957648

Title	Console	Price	Logistic Probability
Game Boy Advance			
Mario Kart: Super Circuit	Game Boy Advance	82.99	0.997253
Tony Hawk's Pro Skater 2	Game Boy Advance	34.99	0.988203
Super Mario Advance	Game Boy Advance	79.99	0.981165
Kirby Nightmare in Dream Land	Game Boy Advance	64.99	0.978152
Castlevania: Circle of the Moon	Game Boy Advance	49.99	0.971780
Super Mario Advance 4: Super Mario Bros 3	Game Boy Advance	76.99	0.971013
Sonic Advance 2	Game Boy Advance	24.54	0.965405
Nancy Drew: Message in a Haunted Mansion	Game Boy Advance	79.99	0.963937
Bomberman Tournament - GBA	Game Boy Advance	74.99	0.948584
Advance Wars 2: Black Hole Rising	Game Boy Advance	74.95	0.943974
GameCube			
Mario Party 7	GameCube	85.11	0.993439
Star Fox Assault	GameCube	69.95	0.991363
Wario World	GameCube	79.95	0.990386
Animal Crossing	GameCube	69.95	0.988285
Soul Calibur 2	GameCube	57.95	0.987732
Star Wars Rogue Squadron II: Rogue Leader	GameCube	69.22	0.987125
Super Monkey Ball 2	GameCube	38.95	0.986841
Lord of the Rings: The Two Towers	GameCube	35.95	0.985317
WWE Wrestlemania XIX	GameCube	44.95	0.984893
Mario Party 5	GameCube	77.95	0.983969

Table C.5.3. Content-Based Recommendation System (CBRS) – Highest & Lowest Probabilities of Positive Recommendation (Price > 20)

Title	Console	Price (> 20)	Logistic Probability
Highest			
Mario Kart: Super Circuit	Game Boy Advance	82.99	0.997253
Lego Star Wars II: The Original Trilogy	PlayStation 2	29.95	0.994129
Mario Party 7	GameCube	85.11	0.993439
Mario Kart DS	Nintendo DS	29.54	0.993222
Halo: Combat Evolved	Xbox	39.98	0.993157
CTR: Crash Team Racing	PlayStation	49.97	0.993145
Star Fox Assault	GameCube	69.95	0.991363
New Super Mario Bros	Nintendo DS	32.00	0.990830
Nintendogs Lab & Friends	Nintendo DS	41.94	0.990707
Wario World	GameCube	79.95	0.990386
Lowest			
Left 4 Dead	PC & Mac	21.99	0.551493
High Heat Baseball 2003	PC & Mac	37.44	0.551730
RPG Maker II	PlayStation 2	39.98	0.553188
Lord of the Rings: Fellowship of the Ring	PlayStation 2	29.98	0.555823
Learn to Speak French 8.0	PC & Mac	53.50	0.560733
Family Guy	PlayStation 2	42.98	0.561059
FIFA Soccer 2003	PlayStation 2	22.97	0.566535
WWE Day of Reckoning 2	GameCube	29.95	0.570980
Champions Return to Arms	PlayStation 2	32.56	0.575103
Doom 3	PC & Mac	49.99	0.579246

Table D.5.3. Content-Based Recommendation System (CBRS) – Highest & Lowest Probabilities of Positive Recommendation (Price < 20)

Title	Console	Price (< 20)	Logistic Probability
Highest			
NBA Street	PlayStation 2	18.82	0.989516
Cooking Mama	Nintendo DS	13.84	0.987274
Age of Empires II, Gold Edition	PC & Mac	11.40	0.986904
Final Fantasy Anthology (Final Fantasy VI)	PlayStation	16.99	0.985707
Sly 2: Band of Thieves	PlayStation 2	13.99	0.984633
Cooking Mama: Cook Off	Nintendo Wii	16.26	0.980485
Tony Hawk's Pro Skater 3	PlayStation 2	16.98	0.979038
Enter the Matrix	GameCube	19.95	0.976942
God of War Chains of Olympus	PlayStation Portable	16.97	0.976161
Ratchet & Clank	PlayStation 2	16.99	0.974414
Lowest			
Resident Evil 4	PC & Mac	11.49	0.550564
Dark Messiah of Might & Magic Elements	Xbox 360	19.45	0.551009
Majestic Chess	PC & Mac	19.95	0.551664
Beyond Atlantis	PC & Mac	15.94	0.552502
Dragonshard	PC & Mac	3.99	0.552548
Space Channel 5	Sega Dreamcast	7.99	0.552844
Icewind Dale 3 in 1 Boxset	PC & Mac	15.97	0.555162
Neverwinter Nights	PC & Mac	3.99	0.557585
The Getaway	PlayStation 2	14.97	0.561097
X Plane	PC & Mac	7.50	0.561413

Table E.5.3. Content-Based Recommendation System (CBRS) – Content-Based Recommendation Examples

Title	Console	Price	Logistic Probability	Cosine Similarity Score
Super Mario 64 DS	**Nintendo DS**	**5 – 30**	–	–
Mario Kart DS	Nintendo DS	29.54	0.993222	0.305531
Nanostray	Nintendo DS	17.95	0.668352	0.191583
Touchmaster	Nintendo DS	13.00	0.963341	0.187996
Meteos	Nintendo DS	23.60	0.764378	0.141974
Castlevania Dawn of Sorrow	Nintendo DS	28.75	0.976882	0.134284
Virtua Tennis 3	**No Filter**	**No Filter**	–	–
Virtua Tennis	Sega Dreamcast	19.98	0.680517	0.485578
Blazing Angels 2: Secret Missions of WWII	PlayStation 3	17.49	0.823615	0.252728
Mario Power Tennis	GameCube	58.49	0.926555	0.222006
Table Tennis	Xbox 360	8.99	0.820792	0.167205
Virtua Fighter 5	PlayStation 3	15.90	0.696892	0.160936
Grand Theft Auto 3	**PlayStation 2**	**5 – 30**	–	–
Greatest Hits: Grand Theft Auto III	PlayStation 2	7.98	0.910993	0.525864
True Crime Streets of LA	PlayStation 2	22.98	0.661429	0.193973
Grand Theft Auto: The Trilogy	PlayStation 2	21.50	0.820799	0.187680
Grand Theft Auto: Vice City	PlayStation 2	8.98	0.876127	0.181681
The Getaway	PlayStation 2	14.97	0.561097	0.109644
Rollercoaster Tycoon 3	**PC & Mac**	**5 – 30**	–	–
Rollercoaster Tycoon 3: Gold Compilation Pack	PC & Mac	19.61	0.574287	0.653565
Rollercoaster Tycoon Gold Edition	PC & Mac	23.95	0.655515	0.394297
Rollercoaster Tycoon Expansion Pack: Loopy Landscapes	PC & Mac	9.95	0.791723	0.379137
Rollercoaster Tycoon 3: Soaked! Expansion	PC & Mac	13.99	0.578184	0.375407
Rollercoaster Tycoon Expansion Pack: Corkscrew Follies	PC & Mac	6.95	0.850402	0.361904
Battlefield: Vietnam	**No Filter**	**No Filter**	–	–
Battlefield 2: The Complete Collection DVD	PC & Mac	5.99	0.655806	0.210539
Battlefield 2: Modern Combat	Xbox 360	22.94	0.834916	0.20294
Battlefield 1942: Deluxe Edition	PC & Mac	32.36	0.892963	0.195767
Shell Shock: Nam '67	PlayStation 2	33.97	0.741913	0.194120
Joint Operations: Typhoon Rising	PC & Mac	11.10	0.616009	0.187984